Supplementary Material

Symptoms of Protracted Alcohol Withdrawal in Patients with Alcohol Use Disorder: A Comprehensive Systematic Review

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Supplementary Table 1. Search string used in various libraries.

Source	Date	String	Filters	N ref
Pubmed	25/2/2020	(craving OR insomnia OR anhedonia OR sleep) (alcohol*) (dependen* OR alcoholism) (withdrawal OR abstinence OR cessation)	From 2006/01/01 English	744
Embase	25/2/2020	(craving:ab,ti OR insomnia:ab,ti OR anhedonia:ab,ti OR sleep:ab,ti) AND alcohol*:ab,ti AND (dependen*:ab,ti OR alcoholism:ab,ti) AND (withdrawal:ab,ti OR abstinence:ab,ti OR cessation:ab,ti) AND [2006-2020]/py NOT [medline]/lim AND [english]/lim AND (article:it OR review:it)	From 2006/01/01 English	63
CDSR	25/2/2020	(craving OR insomnia OR anhedonia OR sleep) (alcohol*) (dependen* OR alcoholism) (withdrawal OR abstinence OR cessation) in Abstract OR (craving OR insomnia OR anhedonia OR sleep) (alcohol*) (dependen* OR alcoholism) (withdrawal OR abstinence OR cessation) in Record Title - (Word variations have been searched)	From 2006/01/01 English	13
		Duplicates		-34
Total	25/2/2020			786

Supplementary Table 2. Description of eligible publications, according to the study design, characteristics of patients.

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
Addolorato, 2007 [1]	Randomized controlled trial	Italy (2003-2006)	Alcohol dependent patients with liver cirrhosis consecutively recruited at the clinic	DSM-IV	12 weeks	• Baclofen, 30 mg/d (n=42) • Placebo (n=42)	C
Anton, 2009 [2]	Randomized double-blind controlled trial	USA (not specified)	Alcohol dependent subjects recruited through advertisements or by phone and abstinent at least 72 h before randomization	DSM-IV	6 weeks	• Flumazenil, 2 mg/d + Gabapentin, max 1200 mg/d (n=33) • Placebo (n=27)	S
Anton, 2011 [3]	Randomized double-blind controlled trial	USA (not specified)	Alcohol dependent patients who had to abstain from alcohol for at least four consecutive days before randomization	DSM-IV	16 weeks	• Naltrexone, 50mg/d (n=50) • Naltrexone, 50 mg/d + Gabapentin, max 1200 mg/d (for the first 6 weeks) (n=50) • Placebo (n=50)	C, S
Baltieri, 2008 [4]	Randomized double-blind controlled trial	Brazil (2005-2007)	Male alcohol dependent outpatients of the hospital	ICD-10	12 weeks	• Topiramate, 300 mg/d (n=52) • Naltrexone, 50 mg/d (n=49) • Placebo (n=54)	C
Beraha, 2016 [5]	Multicenter, randomized placebo-controlled trial	Netherlands (not specified)	Alcohol dependents patients recruited from five inpatient or outpatient treatment centers, randomized after detoxification	DSM-IV	16 weeks	• Baclofen, max 150 mg/d (n=58) • Baclofen, 30 mg/d (n=31) • Placebo (n=62)	C
Blodgett, 2014 [6]	Meta-analysis of 7 randomized double-blind controlled trial	Various countries (not specified)	Alcohol dependent patients	Not specified	Max 4 months	• Topiramate, max 300 mg • Placebo	C
Bogenschutz, 2015 [7]	Single-group cohort study	USA (not specified)	Alcohol dependent patients recruited from the community using advertisements in local media and flyers	DSM-IV	36 weeks	• Psilocybin (n=10) administered at week 4 and week 8	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
Bogenschutz, 2016 [8]	Open-label pilot study	USA (not specified)	Alcohol dependent patients with anxiety disorder symptoms, recruited from an outpatient addiction treatment program and from the community through advertisements	DSM-IV	28 weeks	Lorazepam, max 6 mg/d + Disulfiram, max 500 mg/d (n=41) • Psychosocial treatment	C
Brower, 2008 [9]	Randomized double-blind pilot study	USA (not specified)	Alcohol dependent outpatients from alcohol treatment center or from surrounding community by advertisement, having problems of insomnia not related to alcohol intoxication or withdrawal	DSM-IV	12 weeks	• Gabapentin 1500 mg/d (n=10) • Placebo (n=11)	S
Caputo, 2011 [10]	Cohort study	Italy (not specified)	Alcohol dependent patients consecutively admitted at the hospital who completed detoxification treatment	DSM-IV-TR	12 weeks	• Sodium oxybate, 50 mg/kg (n=48)	C
Caputo, 2014 [11]	Cohort study	Italy (not specified)	Alcohol dependent consecutive patients recruited at three outpatients clinics, who completed detoxification treatment for alcohol dependence	DSM-IV-TR	12 weeks	• Sodium oxybate, 50 mg/kg (n=48)	C
Chakravorty, 2014 [12]	Randomized double-blind placebo-controlled trial	USA (not specified)	Alcohol dependent patients who complained of sleep-related disturbance and recruited via provider referrals from the clinic or through posted flyers	MINI	8 weeks	• Quetiapine, max 400 mg/d (n=10) • Placebo (n=10)	S
Chakravorty, 2019 [13]	Randomized controlled pilot study	USA (not specified)	Veterans alcohol dependent subjects with insomnia symptoms recruited through flyers	MINI	6 months	• CMB-I (n=11) • Monitor only (n=11)	C, S

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
			posted within the medical center ot as direct referrals from their care providers				
Collins, 2015 [14]	Single-arm pilot study	USA (not specified)	Homeless alcohol dependent patients recruited from two community-based agencies for homeless people	DSM-IV-TR	12 weeks	Naltrexone (n=31) • Harm reduction counselling and medication management	C
Cordovil De Sousa Uva, 2010 [15]	Cohort study	Belgium (not specified)	Alcohol dependent patients recruited during a detoxification and rehabilitation program	DSM-IV	18 days	• Diazepan, max 40 mg/d (n=35)	C
de Bejczy, 2015 [16]	Double blind randomized clinical study	Sweden (2009-2011)	Alcohol dependent patients recruited via advertisements in local newspapers	DSM-IV-TR	12 weeks	• Varenicline, max 2mg/d (n=51) • Placebo (n=64)	C
De Sousa, 2008 [17]	Randomized open label not blinded clinical trial	India (not specified)	Male alcohol dependent patients undergoing inpatient detoxification in a clinic	DSM-IV	9 months	• Disulfiram, 250 mg/d (n=50) • Topiramate, 150 mg/d (n=50)	C
De Sousa, 2014 [18]	Randomized not blinded trial	India (not specified)	Adolescents with alcohol dependence undergoing out-patient detoxification	Not specified	6 months	• Disulfiram, max 500 mg/d (n=29) • Naltrexone, 100 mg/d (n=29)	C
De Timary, 2013 [19]	Cohort study	Belgium (not specified)	Alcohol dependent inpatients recruited by three psychiatrists from a withdrawal and detoxification program	DSM-IV	14-18 days	• Diazepam, max 40 mg/d (n=30)	C
Den Uyl, 2017 [20]	Randomized blinded trial	Germany (2004)	Alcohol dependent inpatients	Not specified	3 months	• Active tDCS + CBM (n=20) • Sham tDCS + CBM (n=21) • Active tDCS separate from CBM (n=21)	C
Di Nicola, 2010 [21]	Open prospective	Italy	Alcohol dependent	DSM-IV	14 days	Pregabalin, max 450	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
	study	(2006-2008)	patients with mild-to-moderate AWS who referred to the hospital alcohol unit treatment, undergoing outpatient alcohol detoxification			mg/d (n=40) • Supportive self-help group run by counsellors or psychologists	
Florez, 2008 [22]	Randomized open label trial	Spain (2005-2006)	Alcohol dependent patients who came to the outpatient clinic seeking treatment for alcohol related problems	ICD-10	6 months	• Topiramate, max 400 mg/d (n=51) • Naltrexone, 50 mg/d (n=51)	C
Friedmann, 2008 [23]	Randomized double-blind controlled trial	USA (2002-2006)	Alcohol dependent patients with sleep disturbance during previous periods of abstinence	DSM-IV	6 months	• Trazodone, max 150 mg/d (n=88) • Placebo (n=85)	S
Fucito, 2011 [24]	Randomized clinical trial	USA (2008-2010)	Heavy drinkers seeking smoking cessation treatment recruited through newspaper or internet-based advertisements	Not specified	8 weeks	• Varenicline (active pretratment+active treatment), 2 mg/d (n=15) • Placebo (placebo pretreatment+active treatment) (n=15)	C
Furieri, 2007 [25]	Randomized double-blind controlled trial	Brazil (2004-2005)	Alcohol-dependent outpatients recruited in a public drug treatment center, randomized after detoxification treatment	DSM-IV	4 weeks	• Gabapentin, 600 mg/d (n=30) • Placebo (n=30)	C
Garbutt, 2010 [26]	Randomized double-blind controlled trial	USA (not specified)	Alcohol dependent subjects recruited through newspaper and radio advertisements who were able to refrain from alcohol for 3 days prior to randomization	DSM-IV	12 weeks	• Baclofen, 30 mg/d (n=40) • Placebo (n=40)	C
Gleissenthall, 2014 [27]	Cohort study	Austria (not specified)	Alcohol dependent in-patients admitted for post-withdrawal	ICD-10	8 weeks	• Medical management + psychoteraphy +	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
			treatment			physical exercise (n=54)	
Green, 2019 [28]	Randomized double-blind controlled trial	USA (not specified)	Heavy drinking smokers of East Asian descent	AUDIT	5 days	• Naltrexone, 50 mg/d • Placebo	C
Gupta, 2017 [29]	Randomized open-label trial	India (2012-2013)	Alcohol dependent outpatients presented voluntarily at the department for alcohol detoxification and who were able to maintain sobriety at least 3 days before study entry	ICD-10	12 weeks	• Baclofen, 30 mg/d (n=80) • Benfothiamine (n=75)	C
Hallgren, 2018 [30]	Pharmacotherapy clinical trial	USA (not specified)	Alcohol dependent patients recruited using flyers, advertisements in local papers, Craiglist, and clinical referrals	DSM-IV	12 weeks	• Prazosin,16 mg/d (n=78)	C
Hauser, 2017 [31]	Randomized double-blind controlled trial	USA (not specified)	Veterans alcohol dependent patients with chronic HCV diagnosis, recruited at hepatology clinics or via posted IRB-approved advertisements	DSM-IV-TR	12 weeks	• Baclofen, 30 mg/d (n=88) • Placebo (n=92)	C
Heberlein, 2011 [32]	Case-cohort study	Germany (not specified)	Male alcohol dependent subjects admitted at the hospital for detoxification treatment	DSM-IV and ICD-10	14 days	• Clomethiazole + oxcarbazepine (n=33)	C
Heberlein, 2016 [33]	Case-cohort study	Germany (not specified)	Male alcohol dependent subjects admitted at the hospital for detoxification treatment	DSM-IV and ICD-10	7 days	• Carbamazepine + clomethiazole (n=99)	C
Hillemacher, 2006 [34]	Prospective study	Germany (not specified)	Smoking alcohol dependent patients or early abstainers hospitalized for detoxification	ICD-10	7 days	• Clomethiazole, max 2549 mg/d + carbamazepine, max 900 mg/d (n=127)	C
Huang, 2014 [35]	Randomized double-blind	Taiwan (not specified)	Alcohol dependent subjects recruited from	DSM-IV	7 days	• Dextromethorphan, 360 mg/d (n=18)	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
	placebo-controlled trial		the detoxification and treatment center and had their last drink 24 hours before admission			• Placebo (n=22)	
Jose, 2019 [36]	Open-label randomized trial	India (2019)	Alcohol dependent patients who visited the hospital for alcohol-related problems and completed 1 week of detoxification	ICD-10 DCR	4 weeks	• Baclofen, max 20 mg/d (n=49) • Topiramate, max 100 mg/d (n=45)	C
Kampman, 2007 [37]	Randomized double-blind pilot trial	USA (not specified)	Alcohol dependent outpatients recruited through community referrals and public announcements in the local media who were abstinent from alcohol and free from significant alcohol withdrawal symptoms, as defined by a minimum of 3 consecutive days of abstinence before the start of the study medication.	DSM-IV	12 weeks	• Quetiapine, 400 mg/d (n=29) • Placebo (n=32)	C
Kampman, 2013 [38]	Randomized double-blind controlled trial	USA (not specified)	Cocaine dependent subjects and alcohol users recruited at a treatment center through advertisement in the local media as well as through professional referrals.	Not specified	13 weeks	• Topiramate, 300 mg/d (n=83) • Placebo (n=87)	C
Kharb, 2018 [39]	Short term follow-up study	India (2014-2016)	Male alcohol dependent patients admitted for alcohol detoxification	ICD-10	1 month	• Lorazepam (n=34)	C
Klauss, 2018 [40]	Randomized double-blind controlled trial	Brazil (2015-2018)	Alcohol dependent patients recruited in three specialized clinics	ICD-10 DSM-V	3 months	• tDCS (n=26) • sham tDCS (n=23)	C
Koethe, 2007 [41]	Randomized	Germany	Alcohol dependent	DSM-IV	1 week	• Oxcarbazepine, 900	S, C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
	double-blind controlled pilot trial	(2003-2004)	inpatients seeking treatment for alcohol detoxification			mg/d (n=24) • Placebo (n=26)	
Imbert, 2015 [42]	Cohort study	France (not specified)	Alcohol dependent inpatients treated at the hospital	DSM-IV	3 months	• Baclofen, 180 mg/d (n=67)	C
Lahti, 2011 [43]	Randomized single-blinded trial	Finland (not specified)	Alcohol dependent patients who sought treatment for alcohol withdrawal symptoms at the A-Clinic detoxification center	Not specified	3 days	• Nitrous Oxide (n=32) • Oxygen (n=30) • Air (n=30)	S
Ledda, 2019 [44]	Cohort study	Italy (2012-2016)	Alcohol dependent inpatients consecutively recruited at the hospital at the start of their detoxification treatment	DSM-IV	6 months	• Standardized pharmacological treatment and psychological counselling (n=263)	C
Leggio, 2011 [45]	Retrospective study	Italy (2003-2010)	Alcohol dependent patients who referred to the hospital as outpatients	DSM-IV	Max 42 days	• Metadoxine, max 2000 mg/d (n=94)	C
Leggio, 2015 [46]	Preliminary randomized double-blind placebo-controlled trial	USA (not specified)	Heavy smokers alcohol dependent patients recruited via advertisements in public transportation and mass media, referrals from other clinics and by word-of-mouth	DSM-IV	12 weeks	• Baclofen, 80 mg/d • Placebo	C
Likhitsathian, 2013 [47]	Randomized double-blind controlled trial	Thailand (2010)	Alcohol dependent patients receiving a residential treatment program, recruited at 3 sites, including a psychiatric inpatient unit of a university hospital, a mental health hospital and a drug dependence treatment center.	DSM-IV	12 weeks	• Topiramate, max 300 mg/d (n=53) • Placebo (n=53)	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
Litten, 2013 [48]	Randomized double-blind controlled trial	USA (2011-2012)	Alcohol dependent patients who responded by telephone to advertisements at 5 academic sites	DSM-IV-TR	13 weeks	• Varenicline, max 2 mg/d (n=97) • Placebo (n=101)	C
Malcolm, 2007 [49]	Randomized double-blind controlled comparison trial	USA (not specified)	Alcohol dependent patients with alcohol withdrawal syndrome recruited through referrals from university clinics, emergency departments, community advertisements and word-of-mouth	DSM-IV	12 days	• Gabapentin, max 1200 mg/d • Lorazepam, max 6 mg/d	S
Manning, 2016 [50]	Randomized controlled trial	Australia (2014-2015)	Alcohol dependent inpatients recruited from two residential detoxification facilities	DSM-V	7 days	• CBM training (n=44) • Sham training (n=43)	C
Martinotti, 2007 [51]	Randomized controlled trial	Italy (2005-2006)	Detoxified alcohol dependent patients consecutively recruited from the hospital	DSM-IV	3 months	• Oxcarbazepine, max 900 mg/d (n=28) • Oxcarbazepine, max 1800 mg/d (n=29) • Naltrexone, 50 mg/d (n=27)	C
Martinotti, 2007 [52]	Cohort study	Italy (not specified)	Detoxified alcohol dependent patients who were consecutively recruited	DSM-IV	16 weeks	• Aripiprazole, max 15 mg/d (n=13)	C
Martinotti, 2008 [53]	Cohort study	Italy (2006)	Alcohol dependent patients consecutively recruited at the hospital, who committed the goal of total abstinence and were detoxified before the study period	DSM-IV	16 weeks	Pregabalin, max 450 mg/d (n=20) • Self-help group run by counsellors or psychologists	C
Martinotti, 2008 [54]	Monotherapy open study	Italy (2006)	Alcohol dependent outpatients with a comorbid diagnosis of bipolar disorder,	DSM-IV	16 weeks	• Quetiapine, max 800 mg/d (n=28)	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
			schizoaffective disorder, and/or borderline personality disorder who received detoxification before treatment				
Martinotti, 2009 [55]	Randomized double-blind controlled trial	Italy (2006)	Alcohol dependent patients consecutively recruited at the hospital, receiving detoxification	DSM-IV	16 weeks	• Naltrexone, 50 mg/d (n=28) • Aripiprazole, max 15 mg/d (n=29)	C
Martinotti, 2010 [56]	Multi-centre, randomized, single-blind comparison trial	Italy (2006-2008)	Current alcohol dependent subjects referring to the alcohol treatment unit of the hospital	DSM-IV-TR	14 days	• Pregabalin, max 450 mg/d (n=37) • Tiapride, max 800 mg/d (n=37) • Lorazepam, max 10 mg/d (n=37)	C
Martinotti, 2010 [57]	Multi-centre, randomized, double-blind comparison trial	Italy (2006-2007)	Current alcohol dependent subjects recruited from two hospitals, receiving detoxification and who committed the goal of total abstinence	DSM-IV	16 weeks	• Pregabalin, max 450 mg /d (n=31) • Naltrexone, 50 mg/d (n=28)	C
Martinotti, 2011 [58]	Randomized double-blind controlled trial	Italy (2004-2008)	Alcohol dependent subjects with the presence of anhedonic symptoms who referred to the alcohol treatment unit, receiving detoxification	DSM-IV-TR	142 days	• Acetyl-L-Carnitine, 3000 mg/d (first 10 days) then and 9000 mg/d (n=23) • Acetyl-L-Carnitine, 1000 mg/d (first 10 days) then 9000 mg/d (n=21) • Placebo (n=20)	A
Martinotti, 2014 [59]	Randomized, single-blind controlled trial	Italy (2009-2012)	Current alcohol dependent subjects recruited from two hospitals, receiving detoxification and who committed the goal of total abstinence	DSM-IV-TR	6 weeks	• Topiramate, 100 mg/d (n=26) • Placebo (n=26)	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
Mason, 2009 [60]	Randomized double-blind controlled trial	USA (not specified)	Alcohol dependent paid volunteers who were not seeking treatment, were recruited through advertisements, and were abstinent from alcohol on the day of the study	DSM-IV	7 days	• Gabapentin, 1200 mg/d • Placebo	C, S
Mason, 2014 [61]	Randomized double-blind controlled trial	USA (2004-2010)	Alcohol dependent volunteers who were seeking treatment, were recruited primarily via IRB-approved print and Internet advertisements and were abstinent at least 3 days prior randomization	DSM-IV	12 weeks	• Gabapentin, 900 mg/d (n=54) • Gabapentin, 1800 mg/d (n=47) • Placebo (n=49)	C, S
Melby, 2019 [62]	Randomized double-blind controlled trial	Norway (2016-2017)	Alcohol dependent patients receiving hospital alcohol detoxification	ICD-10	3 days	• Oxytocin nasal spray (n=20) • Placebo (n=20)	S
Miller, 2017 [63]	Meta-analysis of 9 randomized controlled trials	USA, Belgium, South Africa (not specified)	Alcohol dependent patients with insomnia	Not specified	Max 84 days	Various treatment (quetiapine, gabapentin, trazodone)	S
Minozzi, 2018 [64]	Meta-analysis of 12 randomized controlled trial	Various country (not specified)	Alcohol dependent patients	Not specified	At least 12 weeks	• Baclofen • Placebo	C
Mishra, 2010 [65]	Randomized single-blind controlled trial	India (2008)	Male alcohol dependent subjects	ICD-10	9 months	• Active rTMS (n=30) • Sham rTMS (n=15)	C
Mitchell, 2012 [66]	Randomized double-blind controlled trial	USA (not specified)	Heavy-drinking smokers who were seeking treatment for their smoking and were recruited online and invited to participate in a smoking cessation study	Not specified	12 weeks	• Varenicline, 2 mg/d (n=29) • Placebo (n=35)	C
Morley, 2006 [67]	Multi-centre,	Australia	Alcohol dependent	DSM-IV	12 weeks	• Acamprosate, 1998	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
	randomized, double-blind controlled trial	(not specified)	patients who attended an inpatient detoxification program, out-patient treatment or follow-up or who had responded to live or print advertising			mg/d (n=55) • Naltrexone, 50 mg/d (n=53) • Placebo (n=61)	
Morley, 2014 [68]	Randomized double-blind placebo-controlled trial	Australia (not specified)	Alcohol dependent patients recruited by treating clinicians at the outpatient clinic	DSM-IV	12 weeks	• Baclofen, 30 mg/d (n=14) • Baclofen, 60 mg/d (n=14) • Placebo (n=14)	C
Muhonen, 2008 [69]	Randomized double-blind controlled trial	Finland (not specified)	Alcohol dependent patients with depression who were voluntarily seeking outpatient treatment for alcohol problems at three clinics	DSM-IV-TR	26 weeks	• Memantine, 20 mg/d (n=40) • Escitalopram, 20 mg/d (n=40)	C
Muller, 2015 [70]	Randomized double-blind placebo-controlled trial	Germany (2011-2014)	Alcohol dependent patients recruited from in- and outpatient department of the hospital or by spontaneous referral	DSM-IV-TR and ICD-10	24 weeks	• Baclofen, max 270 mg/d (n=28) • Placebo (n=28)	C
Myrick, 2007 [71]	Randomized double-blind controlled trial	USA (not specified)	Alcohol dependent patients who were not seeking treatment and recruited through newspapers and community advertisements	DSM-IV	8 days	• Gabapentin, 1200 mg/d (n=18) • Placebo (n=17)	C, S
Nava, 2006 [72]	Randomized open study	Italy (2001-2002)	Alcohol dependent outpatients recruited through advertisements, words-of-mouth, and local press release	DSM-IV-TR	12 months	• G-hydroxybutyrate, 50 mg/kg (n=28) • Naltrexone, 50 mg/d (n=27) • Disulfiram, 200 mg/d (n=31)	C
Neu, 2018 [73]	Cohort study	Germany (2012)	Alcohol dependent patients who were seeking treatment,	DSM-IV	Max 16 days	• Qualified detoxification (n=77)	S

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
			recruited in two specialized inpatient units for qualified detoxification, and hospitalized for detoxification				
Paparrigopoulos, 2010 [74]	Randomized open pilot controlled trial	Greece (not specified)	Alcohol dependent inpatients who had contacted the clinic for alcohol detoxification	DSM-IV	6 months	• Tiagabine, max 20 mg/d (n=60) • Placebo (n=60)	C
Paparrigopoulos, 2011 [75]	Open-label randomized controlled trial	Greece (not specified)	Alcohol dependent patients who consecutively contacted the clinic and admitted for inpatient alcohol detoxification	DSM-IV-TR	6 weeks	• Topiramate, max 75 mg/d (n=30) • Placebo (n=60)	C
Pedersen, 2013 [76]	Randomized, double-blind, clinical trial	USA (not specified)	Alcohol dependent inpatients undergoing detoxification with symptom-driven PRN lorazepam	Not specified	3 days	• Intranasal oxytocin, 24 IU/dose (n=7) • Placebo (n=4)	C
Perney, 2012 [77]	Multi-center randomized double-blind controlled trial	USA (not specified)	Alcohol dependent outpatients who were primarly recruited by newspaper advertisements and were abstinent for at least 10 days before randomization	DSM-IV	24 weeks	• Acamprosate, 3000 mg/d, (n=335) • Placebo (n=257)	S
Perney, 2018 [78]	Pooled analysis of 12 randomized double-blind controlled trial	Various countries (not specified)	Alcohol dependent patients	Not specified	4-12 months	• Acamprosate (n=1817) • Placebo (n=1691)	S
Petit, 2017 [79]	Cohort study	Belgium (not specified)	Alcohol dependent patients recruited during a detoxification program at the hospital, who did not relapsed during the stay	DSM-IV	18 days	• Diazepam, 29 mg/d (n=41)	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
Petit, 2017 [80]	Cohort study	Belgium (not specified)	Alcohol dependent subjects recruited at the alcohol departments of the hospital and undergoing detoxification	DSM-IV	18 days	• Diazepam, max 60 mg/d (n=256)	C
Petrakis, 2018 [81]	Randomized double-blind controlled trial	USA (not specified)	Alcohol dependent outpatients who responded to advertisements	DSM-IV	12 weeks	• Mecamylamine, 10 mg/d (n=65) • Placebo (n=63)	C
Ponizovsky, 2015 [82]	Randomized double-blind placebo-controlled trial	Israel (2009-2010)	Alcohol dependent patients recruited from 15 outpatient medical centres	ICD-10	52 weeks	• Baclofen, 50 mg/d (n=32) • Placebo (n=32)	C
Ray, 2011 [83]	Randomized, double-blind, placebo-controlled study	USA (not specified)	Alcohol dependent patients who were not seeking treatment	DSM-IV	6 weeks	• Quetiapine, 400 mg/d (n=10) • Placebo (n=10)	C, S
Reynaud, 2017 [84]	Randomized, double-blind, placebo-controlled study	France (not specified)	Outpatients fully abstinent for 3-14 days before randomization	DSM-IV	6 months	• Baclofen, 180 mg/d (n=99) • Placebo (n=91)	C
Roberts, 2017 [85]	Randomized, double-blind, placebo-controlled study	USA (not specified)	Heavy drinking alcohol dependent volunteers	DSM-IV	7 days	• Varenicline, 1 mg/d (n=20) • Varenicline, 2 mg/d (n=20) • Placebo (n=20)	C
Roberts, 2018 [86]	Double-blind, placebo-controlled human laboratory study	USA (not specified)	Heavy drinking tobacco users	Not specified	7 days	• Varenicline, 2 mg/d (n=9) • Varenicline, 2 mg/d + naltrexone, 25 mg/d (n=11) • Placebo (n=10)	C
Rose, 2018 [87]	Meta-analysis of 12 RCTs	Various countries (not specified)	Alcohol dependent subjects	Not specified	3-26 weeks	• Baclofen • Placebo	C
Rozatkar, 2016 [88]	Cohort study	India (2009-2010)	Alcohol dependent subjects who received treatment in the hospital and were detoxified with	ICD-10	4 weeks	• Multiple drugs (n=113)	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
			benzodiazepine alone or with baclofen				
Rubio, 2009 [89]	Randomized, double-blind, placebo-controlled pilot 1study	Spain (not specified)	Male alcohol dependent subjects recruited from alcoholism treatment units	DSM-IV	12 weeks	• Topiramate (n=31) • Placebo (n=32)	C
Rubio, 2010 [90]	Multicenter randomized double-blind pilot controlled trial	Spain (not specified)	Alcohol dependent patients who were seeking treatment	DSM-IV	3 weeks	• Zonisamide, max 600 mg/d (n=20) • Diazepam, max 130 mg/d (n=20)	C
Schlauch, 2013 [91]	Longitudinal study	USA (not specified)	Alcohol dependent patients seeking outpatient dual diagnosis treatment from a community mental health center with a current schizophrenia-spectrum or bipolar disorder. Participants had BAL of 0 at each in-person interview.	DSM-IV	6 months	• Not specified (n=278)	C
Schmidt, 2011 [92]	Cohort study	Germany (2003)	Alcohol dependent inpatients consecutively recruited at the hospital at the start of their detoxification treatment	DSM-IV and ICD-10	24 months	• Standard detoxification treatment with psychoanalytical and behavioural approaches (n=198)	C
Schneekloth, 2012 [93]	Cohort study	USA (not specified)	All alcohol dependent patients treated in the hospital, completing a minimum of three weeks in the Intensive Addiction Program, a 1-month abstinence-based addiction treatment program with on-site residence.	DSM-IV	1 year	Multiple drugs (n=314) • Educational and cognitive behavioural groups	C
Simioni, 2017 [94]	Randomized pilot	France	Alcohol dependent	DSM-IV-	12 weeks	• Diazepam 10 days,	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
	open label controlled clinical trial	(2014-2016)	outpatients recruited through the outpatient alcohol cessation program	TR		max 40 mg/d (n=14) • Diazepam 30 days, max 40 mg/d (n=12)	
Staner, 2006 [95]	Randomized double-blind placebo-controlled trial	France (not specified)	Male alcohol dependent subjects recruited via advertisements and from the outpatient department of the hospital	DSM-IV	15 days	• Acamprosate, 666 mg/d (n=12) • Placebo (n=12)	S
Steinig, 2011 [96]	Cohort study	Germany (not specified)	Alcohol dependent patients recruited during their stay at the hospital for detoxification	ICD-10	4 weeks	• Not specified (n=37)	S
Stock, 2013 [97]	Randomized double-blind controlled trial	USA (not specified)	Alcohol dependent veteran subjects who referred to the clinic for detoxification	DSM-IV	7 days	• Gabapentin, max 1200 mg/d (n=17) • Chlordiazepoxide, max 100 mg/d (n=9)	C, S
Subbaraman, 2013 [98]	Randomized controlled trial	USA (not specified)	Newly abstinent partecipants	Not specified	16 weeks	• Naltrexone, 100 mg/d (n=209) • CBI (n=236) • Naltrexone, 100 mg/d + CBI (n=213) • Placebo (n=205)	C
Umut, 2017 [99]	Cohort study	Turkey (2013-2014)	Alcohol dependent male inpatients admitted at the hospital for detoxification, recruited after 12-24 hours after last alcohol consumption	DSM-IV-TR	10 days	• Not specified (n=43)	C
Von der Goltz, 2011 [100]	Cohort study	Germany (not specified)	Alcohol dependent inpatients admitted for detoxification to the psychiatric department of the University Hospital	DSM-IV	15 days	• Not specified (n=34)	C
Wilhelm, 2011 [101]	Case-cohort study	Germany (not specified)	Male alcohol dependent patients who underwent a detoxification treatment	ICD-10	14 days	• Not specified (n=99)	C
Zgierska, 2008 [102]	Cohort	USA (2007)	Alcohol dependent patients recruited from	DSM-IV	8 weeks	• Course in "Mindfulness	C

First author, year	Study design	Country (study period)	Population	Diagnosis	Follow-up	Treatments (number of participants)	Outcome measures
			three collaborating community-based Intensive Outpatient Treatment Programs (IOPs)			Meditation Relapse Prevention" ("meditation") (n=19)	

A: anhedonia; C: craving; CBI: Combined Behavioral Intervention; HCV: hepatitis C; IRB: institutional review board; MINI: Mini International Neuropsychiatric Interview; rTMS: repetitive transcranial magnetic stimulation; S: sleep problems; tDCS transcranial direct current stimulation

Supplementary Table 3. Description of all eligible studies according to the measures and estimates of selected outcomes (i.e., craving, sleep disorders and anhedonia).

First author, year	Outcome measure	Measure	Estimates	Other comments
Addolorato, 2007 [1]	Craving (OCDS)	Baseline: mean score (95% CI) Follow-up: F-value for treatment effect	**Baclofen** Baseline: 28·0 (23·0-32·0) **Placebo** Baseline: 25·0 (22·0-29·0) Treatment effect: F=13·7, p-value=0·0004	**Comment:** Baclofen significantly reduced craving scores **Other outcome measures:** 8 measures for blood biomarkers
Anton, 2009 [2]	Amount and quality of sleep (ISI, ESS)	Score	No exact scores are available	**Comment:** People with low AW symptoms had better sleep (ISI) with active drug medication, while those with high AW symptoms had better sleep if they were on placebo (interaction F=6.04, p=0.015). Based on ESS, the drug group had a main effect (F=8.83, p<0.004) on individuals on active medication having lower daytime sleepiness than those on placebo, but there was no interaction between the AW status and the medication group. **Other outcome measures:** 4 measures for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
Anton, 2011 [3]	Craving (OCDS)	Mean score (SD)	**Naltrexone+Gabapentin** Baseline: 24·7 (10·2) **Naltrexone** Baseline: 25·1 (8·4) **Placebo** Baseline: 24·8 (8·4) No exact scores are available at follow-up	**Comment:** There were no significant differences between the groups on the OCDS total score in either phase of the study. There was no significant difference between placebo and naltrexone-alone groups but the naltrexone/gabapentin group showed a significantly lower score, i.e. more control over drinking urges, than the naltrexone alone group (t= 2.16, DF 139, p=0.04) and somewhat better than the placebo group (t= 1.69, DF 139, p=0.09).

First author, year	Outcome measure	Measure	Estimates	Other comments
	Amount and quality of sleep (ISI, ESS)	Mean score (SD)	**ISI** **Naltrexone+Gabapentin** Baseline: 10·1 (6·0) **Naltrexone** Baseline: 8·3 (6·2) **Placebo** Baseline: 9·7 (6·6) No exact scores are available at follow-up	**Comment:** Overall, there was a marginally significant difference between the groups over the course of the study (F=2.66, DF 2,140 p=0.07) but during the first phase of the study, while there was no significant difference in reported sleep between the placebo and naltrexone-alone group, the naltrexone/gabapentin group reported significantly better sleep than either the placebo alone group (t=2.49, DF 140, p=0.02) or the naltrexone-alone group (t=2.49, DF 140, p=0.03). Poor sleep quality (high ISI) was significantly related to heavy drinking, but only in the naltrexone-alone group (B=0.261 (0.12), t (139) =2.13, p=0.035). That is, in the naltrexone-alone group, but not in the other two groups, individual subjects were more likely to drink heavily during periods in which they reported poor sleep.
				Other outcome measures: 3 measures for alcohol abstinence 2 measures for blood biomarkers of drinking Mood state (POMS)
Baltieri, 2008 [4]	Craving (OCDS)	Mean score (SD)	**Topiramate** Baseline: 50·7 (13·5) Week 12: 22·4 (9·4) **Naltrexone** Baseline: 50·9 (13·6) Week 12: 22·4 (10·3) **Placebo** Baseline: 47·9 (12·6) Week 12: 21·9 (8·6)	**Other outcome measures:** 5 measures for alcohol abstinence 1 measure for blood biomarkers
Beraha, 2016 [5]	Craving (OCDS)	Mean score (SD)	**Baclofen, 150 mg/d** Baseline: 28·2 (9·2) Week 4: 13·6 (5·8)	**Other outcome measures:** Anxiety (STAI) Depression (BDI)

First author, year	Outcome measure	Measure	Estimates	Other comments
			Week 16: 12·4 (4·9) **Baclofen, 30 mg/d** Baseline: 29·5 (11·1) Week 4: 13·6 (5·8) Week 16: 12·4 (6·4) **Placebo** Baseline: 30·4 (10·1) Week 4: 13·4 (5·1) Week 16: 12·2 (4·3)	2 measures for alcohol abstinence 1 measure for blood biomarkers
Blodgett, 2014 [6]	Craving (OCDS, VAS, PACS, Craving Scale)	Weighted mean score (95% CI)	**Topiramate vs Placebo** Pooled effect: 0·312 (-0·042 to 0·666) p-value: 0·01	**Other outcome measures:** 2 measures for alcohol abstinence 1 measure for blood biomarkers
Bogenschutz, 2015 [7]	Craving (PACS)	Mean scores (SD)	Baseline: 16·00 (5·59) Week 4: 14·10 (7·17) Week 5: 11·89 (8·64) Week 8: 11·56 (5·85) Week 9: 10·00 (6·61) Week 12: 12·11 (8·28) Week 24: 13·00 (9·59) Week 36: 8·11 (9·16)	**Other outcome measures:** Mood state (POMS) 3 measures for psychosocial assessments
Bogenschutz, 2016 [8]	Craving (PACS)	Mean scores (SD)	Baseline: 19·56 (8·05) Week 1: 10·26 (7·19) Week 2: 9·64 (7·15) Week 4: 9·53 (7·60) Week 6: 7·61 (7·20) Week 8: 6·52 (6·94) Week 10: 5·25 (5·32) Week 12: 5·17 (7·09) Week 16: 6·59 (8·12) Week 28: 11·10 (10·22)	**Other outcome measures:** Anxiety (HAM-A) Depression (HAM-D) 2 measures for alcohol dependence
Brower, 2008 [9]	Sleep Problems Questionnaire (SPQ)	Median score (IQR)	**Gabapentin** Baseline: 14·5 (12·0-17·8) 6 weeks: 8·0 (2·5-12·0) 12 weeks: 9·0 (7·0-14·0) **Placebo** Baseline: 18·0 (5·0-19·0) 6 weeks: 8·0 (2·5-11·5) 12 weeks: 2·5 (0·3-7·8)	**Comment**: Also self-reported information on sleep were available. Gabapentin had no differential effects on sleep as measured by either subjective report or measured with polysomnography **Other outcome measures**: 1 measure for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
	Total Sleep Time (TST) measured with polysomnography	Median minutes (IQR)	**Gabapentin** Baseline: 314·5 (294·8-390·8) 3 weeks: 310·9 (281·6-371·0) **Placebo** Baseline: 326·4 (295·8-372·9) 3 weeks: 295·5 (234·4-364·7)	
	Sleep onset latency (SOL) measured with polysomnography	Median minutes (IQR)	**Gabapentin** Baseline: 13·8 (6·5-26·1) 3 weeks: 31·0 (20·6-53·6) **Placebo** Baseline: 22·8 (10·9-30·0) 3 weeks: 16·5 (3·9-40·9)	
	Wake time after sleep onset (WASO) measured with polysomnography	Median minutes (IQR)	**Gabapentin** Baseline: 70·5 (48·1-84·5) 3 weeks: 76·3 (54·4-94·5) **Placebo** Baseline: 94·5 (65·6-119·3) 3 weeks: 89·8 (51·4-171·8)	
Caputo, 2011 [10]	Craving (ACS)	Mean scores (SD)	**Abstinent patients:** **Patients without psychiatric comorbidities:** Baseline: 13·0 (1·6) Week 12: 0·7 (1·4) **Patients with psychiatric comorbidities:** Baseline: 9·8 (5·3) Week 12: 0·6 (0·9) **Non-abstinent patients:** **Patients without psychiatric comorbidities:** Baseline: 12·7 (1·7) Week 12: 5·0 (2·7) **Patients with psychiatric comorbidities:** Baseline: 13·4 (1·8) Week 12: 5·2 (2·9)	**Other outcome measures:** 4 measures for blood biomarkers 1 measure for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
Caputo, 2014 [11]	Craving (ACS)	Mean scores (SD)	**Abstinent patients:** **Lesch I typology of alcohol dependence:** Baseline: 10·7 (5·3) Week 12: 1·2 (1·6) **Lesch II typology of alcohol dependence:** Baseline: 10·5 (3·4) Week 12: 1·0 (1·1) **Lesch III typology of alcohol dependence:** Baseline: 11·0 (5·6) Week 12: 0·25 (0·7) **Lesch IV typology of alcohol dependence:** Baseline: 13·0 (1·7) Week 12: 0·0 (0·0) **Non-abstinent patients:** **Lesch I typology of alcohol dependence:** Baseline: 13·6 (0·9) Week 12: 5·0 (2·4) **Lesch II typology of alcohol dependence:** Baseline: 10·0 (5·2) Week 12: 5·0 (1·7) **Lesch III typology of alcohol dependence:** Baseline: 13·0 (2·4) Week 12: 5·2 (4·4) **Lesch IV typology of alcohol dependence:** Baseline: 13·5 (1·4) Week 12: 5·1 (2·5)	**Other outcome measures:** 4 measures for blood biomarkers 1 measure for alcohol abstinence
Chakravorty, 2014 [12]	Wake time after sleep onset (WASO)	Mean minutes (SD)	**Quetiapine** Baseline: 101·07 (81·80) Week 8: 60·71 (71·41) **Placebo** Baseline: 61·17 (31·77) Week 8: 66·99 (60·62)	**Comment:** Information on sleep efficiency, rapid eye movement, percent of slow wave sleep through the night

First author, year	Outcome measure	Measure	Estimates	Other comments
			Treatment differences: p-value=0·03	
	Sleep onset latency (SOL)	Mean minutes (SD)	**Quetiapine** Baseline: 25·42 (14·12) Week 8: 37·11 (40·06) **Placebo** Baseline: 29·04 (29·51) Week 8: 10·81 (9·64) Treatment differences: p-value=0·08	
	Rapid eye movement sleep latency (REM-L)	Mean minutes (SD)	**Quetiapine** Baseline: 108·10 (51·49) Week 8: 115·00 (63·57) **Placebo** Baseline: 92·60 (52·81) Week 8: 88·75 (53·78) Treatment differences: p-value=0·74	
Chakravorty, 2019 [13]	Sleep latency (SL)	Baseline: mean minutes (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 33·9 (20·3) Week 8 (end of treatment):20·6 (-18·2;59·5) Month 3 after treatment: 23·2 (-20·5;67·0) Month 6 after treatment: 25·8 (-22·7;74·4) **CBI-I:** Baseline: 69·0 (65·5) Week 8 (end of treatment): -45·1(-78·0;-12·2) Month 3 after treatment: -36·8 (-75·9;2·3) Month 6 after treatment: -25·4 (-71·9;20·9)	**Other outcome measures:** 3 measures for sleep 1 measure for alcohol abstinence 1 measure for depression 1 measure for anxiety 2 measures for psychiatric evaluation

First author, year	Outcome measure	Measure	Estimates	Other comments
	Insomnia Severity Index (ISI) total score	Baseline: mean scores (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 19·8 (5·6) Week 8 (end of treatment): -8·4 (-16·3;-0·5) Month 3 after treatment: -9·4 (-18·3;-0·5) Month 6 after treatment: -10·5 (-20·4;-0·6) **CBI-I:** Baseline: 18·0 (5·6) Week 8 (end of treatment): -12·7(-15·5;-10·0) Month 3 after treatment: -13·4 (-16·0;-10·9) Month 6 after treatment: -13·9 (-16·3;-11·6)	
	Wake time after sleep onset (WASO)	Baseline: mean minutes (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 33·7 (26·4) Week 8 (end of treatment): -1·7 (-17·8;14·3) Month 3 after treatment: -1·9 (-20·1;16·1) Month 6 after treatment: -2·1 (-22·3;17·9) **CBI-I:** Baseline: 43·6 (28·4) Week 8 (end of treatment): -24·5 (-40·8;-8·3) Month 3 after treatment: -27·6 (-45·9;-9·3) Month 6 after treatment: -30·7 (-51·0;-10·4)	
	Total Sleep Time (TST)	Baseline: mean minutes (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 283·1 (115·3) Week 8 (end of treatment): 45·2 (0·8;89·6) Month 3 after treatment: 50·9 (0·9;100·8) Month 6 after treatment: 56·5 (1·0;112·0)	

First author, year	Outcome measure	Measure	Estimates	Other comments
			CBI-I: Baseline: 289·6 (84·7) Week 8 (end of treatment): 102·4 (57·5;147·3) Month 3 after treatment: 115·2 (64·7;165·7) Month 6 after treatment: 128·0 (71·9;184·1)	
	Epworth Sleepiness Scale (ESS) total score	Baseline: mean scores (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 13·1 (5·6) Week 8 (end of treatment): -0·6 (-2·4;1·0) Month 3 after treatment: -0·7 (-2·7;1·1) Month 6 after treatment: -0·8 (-3·0;1·3) **CBI-I:** Baseline: 9·1 (4·1) Week 8 (end of treatment): -3·1 (-4·9;-1·4) Month 3 after treatment: -3·5 (-5·5;-1·5) Month 6 after treatment: -3·9 (-6·1;-1·7)	
	Craving (PACS)	Baseline: mean scores (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 9·0 (6·3) Week 8 (end of treatment): -0·5 (-3·6;2·6) Month 3 after treatment: -0·6 (-4·1;2·9) Month 6 after treatment: -0·6 (-4·6;3·2) **CBI-I:** Baseline: 6·1 (7·2) Week 8 (end of treatment): -0·7 (-3·9;-2·3) Month 3 after treatment: -0·8 (-4·4;2·6) Month 6 after treatment: -0·9 (-4·9;2·9)	

First author, year	Outcome measure	Measure	Estimates	Other comments
Collins, 2015 [14]	Craving (PACS)	Median scores (IQR)	Baseline: 21·00 (14·00-28·75) Week 12: 14·00 (5·50-24·50) Over time difference: p-value=0·001	**Other outcome measures:** 4 measures for alcohol abstinence 4 measures for blood biomarkers
Cordovil De Sousa Uva, 2010 [15]	Craving (OCDS)	Scores	No exact scores are available	**Comment:** A significant decrease of craving from onset to the end of withdrawal among alcohol-dependent subjects was observed. However, the alcoholics "craving scores" remained higher than controls at the end of the withdrawal cure the alcoholics 'craving scores' remained higher than controls at the end of the withdrawal cure.
de Bejczy, 2015 [16]	Craving (OCDS)	Baseline: mean score (SD) Follow-up: model-based mean score (CI)	**Varenicline** Baseline: 23·7 (0·77) Week 12 IIT: 4·54 (1·16-7·91) Week 12 PP: 5·64 (1·66-9·62) **Placebo** Baseline: 22·3 (0·66) Week 12 IIT: 6·18 (3·04-9·31) Week 12 PP: 7·62 (3·98-11·26) p-value for treatment effect: IIT: p=0·06 PP: p=0·05	**Other outcome measures:** 5 measures for alcohol abstinence 1 measure for nicotine dependence 3 measures for blood biomarkers
De Sousa, 2008 [17]	Craving (craving scale, not specified)	Mean score (SD)	**Disulfiram** Baseline: 56 (16) Month 9: 12·7 (9·3) **Topiramate** Baseline: 53 (17) Month 9: 18·2 (9·9) Treatment differences at 9th month: p-value=0·0023	**Other outcome measures:** 3 measures for alcohol abstinence 1 measure for blood biomarkers
De Sousa, 2014 [18]	Craving (A-OCDS)	Mean score (SD)	**Disulfiram** Baseline:11·8 (4·3) Month 6: 9·2 (3·1) **Naltrexone** Baseline:11·4 (4·1) Month 6: 5·3 (1·8)	**Other outcome measures:** 1 measure for blood biomarker 2 measures for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
			p-value differences treatment: p=0·0001	
De Timary, 2013 [19]	Craving (OCDS)	Scores	No exact scores are available	**Comment:** Obsessional and compulsional subscores of craving decreased significantly (t=-4.498, p<0.0001) from the beginning to the end of withdrawal **Other outcome measures:** Depression (BDI-II) Self-consciousness state (RSCS)
Den Uyl, 2014 [20]	Craving (PACS)	Mean score (SD)	**Active tDCS +CBM** Pre-treatment: 7·1 (1·0) Post-treatment: 5·6 (1·0) **Sham tDCS +CBM** Pre-treatment: 5·9 (1·0) Post-treatment: 4·4 (1·0) **Active tDCS + separate CBM** Pre-treatment: 4·4 (1·0) Post-treatment: 4·1 (1·0)	
Di Nicola, 2010 [21]	Craving (VASc)	Mean scores	**Patients who remained alcohol free** Day 1: 3·5 Day 2: 2·3 Day 3: 1·5 Day 4: 1·0 Day 5: 0·7 Day 6: 0·5 Day 7: 0·3 Day 14: 0·6	**Comment:** A significant reduction of craving for alcohol during the 7 days of outpatient detoxification has been observed.
	Craving (OCDS)	Scores	No exact scores are available	**Comment:** Abstinent patients showed a significant reduction between first and last day of evaluation (day 1 and 14) for the OCDS (Z=-4.374, p<0.001), and its subscores of OCDS obsessive (Z=-4.194, p<0.001) and OCDS compulsive (Z=-4.378, p<0.001).
Florez, 2008 [22]	Craving (OCDS)	Mean score (SD)	**Naltrexone** Baseline: 18·33 (7·064) Month 3: 5·65 (8·492)	**Other outcome measures:** Quality of life (EQ-5D) 3 measures for alcohol dependence

First author, year	Outcome measure	Measure	Estimates	Other comments
			Month 6: 5·11 (6·846) **Topiramate** Baseline: 16·04 (7·150) Month 3: 2·75 (3·928) Month 6: 2·72 (4·595) Intergroup comparison baseline-follow-up: p-value=0·629 Intergroup comparison 3 to 6 months: 0·910	1 measure for nicotine dependence 5 measures for blood biomarkers
Friedmann, 2008 [23]	Sleep quality (PSQI)	Scores, slope for mixed linear regression (95% CI)	No exact scores are available· Linear slope: -3·43 (−4·18, −2·68)	**Comment:** Sleep quality improved more for the trazodone group than the placebo group at the 1 and 3 month follow-ups, with a moderate effect size from baseline to 1 month assessment (Cohen's d, -0.50) and a large effect from baseline to the 3 month follow-up (Cohen's d, -0.93). However, sleep quality degraded for the trazodone group between the 3 and 6 month assessments; such that sleep quality was equal for both groups at the 6-month assessment. **Other outcome measures:** 3 measures for alcohol abstinence Health status (SF-12), mental health and physical health components
Fucito, 2011 [24]	Craving (OCDS)	Mean score (SD)	**Varenicline** Baseline: 7·95 (3·71) Week 8 change: 2·43 (3·16) **Placebo** Baseline: 8·16 (3·43) Week 8 change: 1·57 (3·57) p-value=0·03	**Other outcome measures:** 1 measure for alcohol abstinence 2 measures for smoking dependence
Furieri, 2007 [25]	Craving (OCDS)	Scores	No exact scores are available·	**Comment:** OCDS was stratified by 4 groups of items. All parameters were significantly reduced (p<0.01) at weeks 2 and 4 in both groups compared to baseline. The

First author, year	Outcome measure	Measure	Estimates	Other comments
				gabapentin group showed lower mean scores than the placebo group, however this difference was significant only for automaticity of drinking (p=0.0475). The mean score of items defining “actual craving” decreased significantly at weeks 2 and 4 in both groups compared to baseline. This decrease was greater for the gabapentin group, since the mean scores were significantly lower (p<0.01) in this group than in the placebo group at both weeks 2 and 4. **Other outcome measures:** 6 measures for blood biomarkers 4 measures for alcohol abstinence
Garbutt, 2010 [26]	Craving (PACS)	Mean (SD)	**Baclofen** Baseline: 14·4 (5·8) Week 12: 11·6 (5·9) **Placebo** Baseline: 16·4 (5·6) Week 12: 13·5 (6·3) p-value treatment differences=0·13	**Other outcome measures:** Anxiety (STAI) Depression (SDS) 3 measures for alcohol abstinence
Gleissenthall, 2014 [27]	Craving (OCDS)	Mean scores (SD)	Week 1: 16·7 (7·7) Week 8: 9·1 (6·4)	**Other outcome measures:** Depression (MSDRS and BDI) 12 measures for blood biomarkers
Green, 2019 [28]	Craving (UF)	Scores	No exact scores are available	**Comment:** There was a significant main effect of medication (b=-0.45, SE=0.19, p=0.02) with participants having less urge for alcohol while on Naltrexone in comparison to placebo. **Other outcome measures:** Craving for smoking
Gupta, 2017 [29]	Craving (OCDS)	Mean score (SD)	**Baclofen** Baseline: 27·63 (7·42) Decrease week 12: 16·56 (10·5) **Benfothiamine**	**Other outcome measures:** Anxiety (HARS) Depression (HDRS) 3 measures for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
			Baseline: 26·22 (8·31) Decrease Week 12: 7·48 (9·75) Treatment difference at baseline: p-value=0·328 Treatment difference at week 12: p-value=0·074	2 measures for blood biomarkers
Hallgren, 2018 [30]	Craving (PACS)	Multivariate mean scores (SE)	**Abstinence initiators:** Average craving: 2·73 (0·45), p<0·001 Post-abstinence immediate change: -0·82 (0·28), p-value=0·004 Post-abstinence change over time (continuous, per week): -0·21 (0·04), p-value=<0·001 **Already abstainers:** Average craving: 0·98 (0·22), p-value<0·001 Change over time: -0·04 (0·02), p-value=0·043 **Continued drinkers:** Average craving: 4·61 (0·23), p-value<0·001 Change over time: -0·10 (0·05), p-value=0·033	**Other outcome measures**: Depression (PHQ-9)
Hauser, 2017 [31]	Craving (OCDS)	Mean score (95% CI)	**Baclofen** Baseline: 20·55 (18·51-22·58) Week 6: 10·65 (8·58-12·73) Week 12: 8·79 (6·71-10·86) **Placebo** Baseline: 19·50 (17·51-21·49) Week 6: 12·13 (10·11-14·14) Week 12: 9·52 (7·51-11·54) Mean difference between groups at 12 weeks -0·74 (-3·63 to 2·15) p-value: 0·62	Mean scores of resistance, interference, and obsessive components of OCDS are also available. **Other outcome measures:** Anxiety (BSI) Depression (BDI-II) Post-traumatic stress disorders (PCL-C) 3 measures for alcohol abstinence
Heberlein, 2011 [32]	Craving (OCDS)	Mean scores (SD)	Day 1: 19·42 (7·39)	**Other outcome measures:**

First author, year	Outcome measure	Measure	Estimates	Other comments
			Day 7: 10·59 (6·99) Day 14: 9·34 (6·73)	Depression (BDI) Anxiety (STAI) 1 measure for blood biomarkers
Heberlein, 2016[33]	Craving (OCDS)	Mean scores (SD)	**Total:** Day 1: 19·80 (6·70) Day 7: 10·92 (7·42) Day 14: 9·88 (6·71) **Early abstinent:** Day 1: 18·42 (6·33) Day 7: 8·58 (7·08) Day 14: 8·90 (5·41) **Positive breath alcohol:** Day 1: 19·81 (6·96) Day 7: 11·43 (7·12) Day 14: 10·54 (7·01)	**Other outcome measures:** Depression (BDI) Anxiety (STAI) 1 measure for alcohol abstinence 3 measures for blood biomarkers
	Craving (PACS)	Mean scores (SD)	**Total:** Day 1: 16·15 (7·56) Day 7: 6·27 (5·78) Day 14: 5·07 (6·17) **Early abstinent:** Day 1: 16·88 (6·30) Day 7: 4·73 (4·52) Day 14: 3·75 (4·09) **Positive breath alcohol:** Day 1: 15·73 (7·75) Day 7: 6·69 (5·95) Day 14: 5·29 (6·48)	
Hillemacher, 2006 [34]	Craving (OCDS)	Mean scores (SD) and median scores (Range)	**Mean scores:** Day 0: 20·7 (7·73) Day 7: 12·3 (6·31) **Median scores:** Day 0: 20·0 (3-36) Day 7: 11·0 (3-33)	**Comment:** Obsessive and Compulsive components of OCDS craving are also provided **Other outcome measures:** 1 measure for alcohol dependence
Huang, 2014 [35]	Craving (OCDS)	Mean score (SD)	**Dextromethorphan** Day 1: 37·5 (10·9) Day 2: 28·7 (11·5) Day 3: 23·8 (11·9) **Placebo** Day 1: 35·4 (8·1) Day 2: 25·5 (8·7)	**Comment:** Obsessive and Compulsive craving are also provided. **Other outcome measures:** 1 measure for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
			Day 3: 20·9 (7·6)	
Jose, 2019 [36]	Craving (OCDS)	Mean score (SD)	**Baclofen** Baseline: 26·52 (5·581) Week 1: 16·64 (10·123) Week 2: 9·21 (7·942) Week 3: 5·48 (7·983) Week 4: 2·0 (7·036) **Topiramate** Baseline: 28·36 (6·769) Week 1: 26·08 (7·410) Week 2: 19·44 (9·747) Week 3: 14·50 (11·359) Week 4: 6·83 (9·855)	Mean scores for obsessive and compulsive components of OCDS are also available.
	Craving (VAS)	Mean score (SD)	**Baclofen** Baseline: 81·20 (8·817) Week 1: 63·20 (17·072) Week 2: 39·56 (14·924) Week 3: 24·78 (23·906) Week 4: 6·19 (18·568) **Topiramate** Baseline: 82·40 (10·116) Week 1: 78·00 (11·547) Week 2: 65·20 (17·823) Week 3: 48·63 (26·603) Week 4: 25·56 (28·122)	
				Other outcome measures: 5 measures for alcohol abstinence
Kampman, 2007 [37]	Craving (PACS)	Scores	No exact scores are available	**Comment:** Craving among Type B alcoholics declined significantly more in quetiapine-treated compared with placebo-treated patients ($\chi 2 = 3.74$, $P = 0.04$). There were no significant differences in craving between quetiapine-treated and the placebo-treated Type A patients, or between either quetiapine or placebo-treated Type A patients and quetiapine-treated Type B patients. **Other outcome measures:**

First author, year	Outcome measure	Measure	Estimates	Other comments
				4 measures for alcohol abstinence Depression (HAM-D) Anxiety (HAM-A)
Kampman, 2013 [38]	Craving (PACS)	Scores	No exact scores are available·	**Comment:** Alcohol craving measured by PACS scores declined significantly over time during the trial (Z= 7.41, p <.001) and scores were significantly lower among topiramate treated subjects (Z= 1.99, P<.05). **Other outcome measures:** 4 measures for cocaine dependence 2 measures for addiction problems and improvement 1 measure for alcohol abstinence
Kharb, 2018 [39]	Craving (PACS)	Mean scores (SD)	**Non-relapsed patients:** At discharge: 7·78 (1·56) 1 month later: 12·44 (4·33) **Relapsed patients:** At discharge: 15·90 (2·51) 1 month later: 22·29 (3·15) Group difference at discharge: t=-4·149, p<0·0001 Group difference at 1 month later: t=-4·014, p<0·0001	
Klauss, 2018 [40]	Craving (OCDS)	Scores	No exact scores are available	**Comment:** The difference between the slopes from both groups was statistically significant [F(1, 6) = 19.19, p = 0.00047], showing that the decrease of OCDS scores was significantly larger in the tDCS group. When comparing craving scores obtained before (initial) and after (final) treatment by paired t-tests, a statistically significant difference t(22) = 4.06, p = 0.0005, 95% CI [1.93, 5.98] was observed for the real tDCS group only (Figure 3D), showing that OCDS scores were significantly smaller than

First author, year	Outcome measure	Measure	Estimates	Other comments
				baseline values after 10 sessions of bilateral cortical DC stimulation
Koethe, 2007 [41]	Sleep quality (PSQI)	Mean scores (SD)	**Oxcarbazepine** Baseline: 6·8 (4·17) Day 6: 6·4 (4·81) **Placebo** Baseline: 9·6 (4·02) Day 6: 7·3 (4·45) Treatment difference at the end of the study: -0·9 p-value: 0·54	**Other outcome measures:** Anxiety (STAI) Depression (BDI) Physical and psychic symptoms (BSI) 1 measure for alcohol dependence
	Craving (VAS)	Mean scores (SD)	**Oxcarbazepine** Baseline: not reported Day 6: 18·2 (20·27) **Placebo** Baseline: not reported Day 6: 13·6 (15·83) Treatment difference at the end of the study: 4·6 p-value: 0·20	
Imbert, 2015 [42]	Craving (OCDS)	Scores	No exact scores are available	**Comment:** Craving scores are shown in a figure, overall and by responders and non-responders to the treatment.
Lahti, 2011 [43]	Total sleeping time (TST)	Median hours (SD)	**Nitrous oxide:** Day 1: 6·9 (0·7) Day 2: 6·6 (0·5) Day3: 6·6 (0·5) **Oxygen:** Day 1: 6·8 (0·3) Day 2: 6·8 (0·5) Day 3: 6·8 (0·5) **Air:** Day 1: 7·1 (0·5) Day 2: 6·6 (0·7) Day 3: 6·8 (0·5)	**Other outcome measures:** 5 measure for sleep parameters
Ledda, 2019 [44]	Craving (OCDS)	Scores	No exact scores are available	**Comment:** A significant decrease was observed for both relapsers and abstainers in the OCDS (Relapsers: F (4,256) = 3.49,

First author, year	Outcome measure	Measure	Estimates	Other comments
				3.69, 2.68; p-value < 0.05) Obsessive and Compulsive components of OCDS craving were also evaluated
Leggio, 2011 [45]	Craving (VAS)	Mean scores (SD)	Before treatment: 5·67 (3·36) After treatment: 2·70 (3·42) p-value: <0·001	**Other outcome measures:** 1 measure for alcohol abstinence 4 measures for blood biomarkers
Leggio, 2015 [46]	Craving (AUQ and OCDS)	Scores	No exact scores are available	**Comment:** Baclofen, as compared with placebo, did not significantly reduced alcohol craving, assessed by either the AUQ or the OCDS. **Other outcome measures:** 2 measures for alcohol craving 1 measure for smoking abstinence 1 measure for alcohol abstinence
Likhitsathian, 2013 [47]	Craving (11 point-Lickert type questionnaire)	Mean score (SD)	**Topiramate** Baseline: 0·6 (1·5) Week 4: 1·8 (2·1) Week 8: 2·0 (2·1) Week 12: 2·4 (2·4) **Placebo** Baseline: 0·9 (1·7) Week 4: 1·7 (2·0) Week 8: 2·0 (2·0) Week 12: 2·0 (2·2) p-value differences baseline: 0·39 p-value differences week 4: 0·84 p-value differences week 8: 0·92 p-value differences week 12: 0·42	**Other outcome measures:** 4 measures for alcohol abstinence 1 measure for blood biomarker Quality of life (SF-36)
Litten, 2013 [48]	Craving (PACS)	Baseline: mean score (SD) Follow-up: model-based mean score (95% CI)	**Varenicline** Baseline: 17·7 (6·2) Weeks 6, 10, 14: 9·9 (8·9-10·9) **Placebo** Baseline: 16·7 (6·8) Weeks 6, 10, 14: 11·6 (10·7-12·6)	**Other outcome measures:** Quality of life (SF-12) 7 measures for alcohol abstinence 1 measure for smoking dependence

First author, year	Outcome measure	Measure	Estimates	Other comments
			p-value differences baseline: 0·276 follow-up mean difference: 1·7 p-value: 0·011	
Malcolm, 2007 [49]	Daytime sleepiness (ESS)	Scores	No exact scores are available	**Comment:** An overall main effect of drug on sleepiness during treatment was noted in the analysis (F=4.42, p=0.0394) with patients receiving lorazepam reporting significantly more sleepiness than those receiving gabapentin.
	Quality of Sleep (BDI-Item 16)	Scores	No exact scores are available	**Comment:** Overall, lorazepam and gabapentin were equally efficacious in reducing this measure of sleep difficulty during treatment, evidenced as a nonsignificant drug effect during treatment (p>0.14). However, during the follow-up period, significant differences were noted by prior drug treatment across the period (F=5.02, p=0.029) While discontinuation of gabapentin did not alter this measure, patients previously treated with lorazepam reported increased sleep difficulty, an effect which dissipate by the end of the study.
	Insomnia (CIWA-AR-Item 11)	Scores	No exact scores are available	**Comment:** On the first day of the treatment, self-reports of insomnia were significantly lower in patients receiving lorazepam in comparison to gabapentin (F=7.48, p=0.008). In contrast, in the follow-up period, patients previously receiving lorazepam experienced increased insomnia, while gabapentin patients continued to report a reduction in insomnia. These impressions were supported by statistical analysis indicating both a drug by follow-up day

First author, year	Outcome measure	Measure	Estimates	Other comments
				effect (F=4.82, p=0.032), as well as a significant difference on the first follow-up day in self-reports of insomnia with subjects previously receiving lorazepam reporting higher levels of insomnia in comparison to prior gabapentin treatment (F=4.07, p=0.048).
Manning, 2016 [50]	Craving (ACQ-SF-R)	Scores	No exact scores are available	**Comment:** A significant main effect of time on craving was observed, F(1, 62) = 32.83, p < 0.001, g2 = 0.35, with lower scores reported post-training relative to baseline among participants with craving data available at both time points (n = 32 in each group).However, there was no significant main effect of group, F(1, 62) = 2.70, p = 0.11, g2 = 0.04, nor a significant interaction effect between group and time, F(1, 62) = 3.01, p = 0.09, g2 = 0.05.
Martinotti, 2007 [51]	Craving (OCDS)	Mean score (SD)	**Oxcarbazepine, 900 mg** Baseline: 16·1 (10·2) Day 45: 9·5 (7·2) Day 90: 7·2 (7·4) **Oxcarbazepine, 1800 mg** Baseline: 22·9 (13·3) Day 45: 13·0 (10·4) Day 90: 6·2 (8·8) **Naltrexone** Baseline: 26·8 (9·7) Day 45: 7·1 (6·0) Day 90: 6·8 (7·7)	Mean scores for obsessive and compulsive components of OCDS are also available. **Other outcome measures:** 1 measure for psychiatric symptomatology 3 measures for alcohol dependence 4 measures for blood biomarkers
	Craving (VAS)	Mean score (SD)	**Oxcarbazepine, 900 mg** Baseline: 2·8 (3·8) Day 45: 3·2 (3·7) Day 90: 1·3 (1·9) **Oxcarbazepine, 1800 mg** Baseline: 3·9 (3·6) Day 45: 2·9 (3·3)	

First author, year	Outcome measure	Measure	Estimates	Other comments
			Day 90: 2·0 (2·4) **Naltrexone** Baseline: 3·4 (3·3) Day 45: 1·8 (1·9) Day 90: 1·1 (1·5)	
Martinotti, 2007 [52]	Craving (VAS)	Mean score reductions from baseline	Baseline: not reported Week 2: -3·04 Week 8: -3·16 Week 16: -3·64 p-value=0·002	
	Craving (OCDS)	Scores	No exact scores are available	**Comment:** A significant progressive reduction was observed (p<0.05)
				Other outcome measures: 4 measures for blood biomarkers Psychiatric symptoms (SCL-90 R)
Martinotti, 2008 [53]	Craving (VAS)	Mean score (SD)	Baseline: 6·0 (2·1) Week 2: 4·1 (2·0) Week 8: 2·5 (2·0) Week 16: 1·2 (1·9)	
	Craving (OCDS)	Scores	No exact scores are available	**Comment:** A significant progressive reduction of craving was evaluated with OCDS (F=19.64; p-value<0.001)
				Other outcome measures: 1 measure for alcohol abstinence Psychiatric symptoms (SCL-90-R)
Martinotti, 2008 [54]	Craving (OCDS)	Mean scores (SD)	Baseline: 18·8 (11·7) Week 16: 10·2 (11·3) Change: 8·06 p-value:0·0008	**Comment:** Obsessive and Compulsive components of OCDS are also provided
	Craving (VAS)	Mean scores (SD)	Baseline: 3·9 (3·3) Week 16: 2·5 (2·3) Change: 1·04 p-value:0·018	
				Other outcome measures: 1 measure for alcohol abstinence 3 measures for psychopathological symptoms
Martinotti, 2009 [55]	Craving (OCDS)	Scores	No exact scores are available	**Comment:** The Naltrexone group showed a significant mean reduction in the OCDS

First author, year	Outcome measure	Measure	Estimates	Other comments
				(F = 35.26; P < 0.001) and its subscores of OCDS obsessive (F = 12.02; P < 0.001) and OCDS compulsive (F = 29.86; P < 0.001).
	Craving (VAS)	Scores	No exact scores are available	**Comment:** Both the Aripiprazole group and the Naltrexone group showed a significant reduction between times for the VAS (F = 6.66; P < 0.05, in the Aripiprazole group and F = 3.27; P < 0.5 in the Naltrexone group).
				Other outcome measures: 4 measures for alcohol abstinence 3 blood biomarkers Psychiatric symptoms (SCL-90-R)
Martinotti, 2010 [56]	Craving (OCDS)	Mean score (SD)	**Lorazepam** Baseline: 20·03 (8·71) Day 14: 10·94 (9·4) Change: p<0·001 **Tiapride** Baseline: 18·1 (7·44) Day 14: 11·16 (9·34) Change: p=0·005 **Pregabalin** Baseline: 19·25 (7·09) Day 14: 8·47 (7·71) Change: p<0·001	Mean change of obsessive and compulsive components of OCDS are also available. **Other outcome measures**: Quality of life (QL-index) Psychiatric symptoms (SCL-90-R) **Comment**: Significant reduction between baseline and the end of the treatment was found in all the groups at the OCDS and the VASc, at the SCL-90-R and QL-index (p<0.001).
	Craving (VASc)	Score	Data shown in figure without exact values	
Martinotti, 2010 [57]	Craving (OCDS)	Mean score (SD)	**Pregabalin** Baseline: 22·1 (10·2) Week 16: 5·5 (7·2) **Naltrexone** Baseline: 26·8 (9·7) Week 16: 7·1 (6·0) Over time p-value pregabalin: <0·001 Over time p-value naltrexone: <0·001	**Comment:** Information on obsessive and compulsive components of OCDS are also available. **Other outcome measures:** 3 measures for alcohol abstinence General symptoms (SCL-90-R GSI) Psychiatric symptoms (SCL-90-R) Phobic anxiety symptoms (SLC-90-R) Quality of life (QL-index)

First author, year	Outcome measure	Measure	Estimates	Other comments
	Craving (VAS)	Mean score (SD)	**Pregabalin** Baseline: 6·0 (1·6) Week 16: 1·3 (1·3) **Naltrexone** Baseline: 5·4 (1·7) Week 16: 2·8 (1·9) Over time p-value pregabalin: <0·001 Over time p-value naltrexone: <0·05	
Martinotti, 2011 [58]	Anhedonia (VASa)	Baseline: mean score (SD) Follow-up: change from baseline in rating scales	**Acetyl-L-Carnitine 3g + 3g** Baseline: 4·19 (2·09) Change 10 days: -1·13 Treatment vs placebo difference at 10 days: −2·6 95% CI: (−4·8 to −0·4) p-value= 0·024 Change 80 days: -1·52 **Acetyl-L-Carnitine 1g + 3g** Baseline: 6·33 (2·06) Change 10 days: -3·20 Treatment vs placebo difference at 10 days: −2·4 95% CI: (−4·4 to −0·5) p-value= 0·017 Change 80 days: -4·28 **Placebo** Baseline: 6·23 (2·26) Change 10 days: -0·60 Change 80 days: -2·53	**Other outcome measures:** Melancholic symptoms (SANS, BRMS)
	Anhedonia (SHAPS)	Baseline: mean score (SD) Follow-up: change from baseline in rating scales	**Acetyl-L-Carnitine 3g + 3g** Baseline: 4·57 (1·77) Change 10 days: -2·71 Treatment vs placebo difference at 10 days: −3·3 95% CI: (−5·1 to −1·5) p-value= 0·001 Change 80 days: -2·57 **Acetyl-L-Carnitine 1g + 3g**	

First author, year	Outcome measure	Measure	Estimates	Other comments
			Baseline: 5·60 (2·10) Change 10 days: -2·80 Treatment vs placebo difference at 10 days: −2·00 95% CI: (−3·6 to −0·4) p-value= 0·016 Change 80 days: -3·50 **Placebo** Baseline: 5·78 (2·12) Change 10 days: -1·00 Change 80 days: -3·67	
Martinotti, 2014 [59]	Craving (OCDS)	Mean score (SD)	**Topiramate** Baseline: 15·5 (9·2) Week 6: 0·5 (0·6) Over time p-value: <0·001 **Placebo** Baseline: 12·7 (6·2) Week 6: 5·0 (4·6) Over time p-value: <0·05	**Other outcome measures:** 4 measures for alcohol abstinence Obsessive-compulsive symptoms (SCL-90-R) Psychiatric symptoms (SCL-90-R, GSI) Quality of life (QL-index)
	Craving (VAS)	Mean score (SD)	**Topiramate** Baseline: 3·4 (1·6) Week 6: 0·3 (0·1) Over time p-value: <0·05 **Placebo** Baseline: 3·9 (1·7) Week 6: 1·0 (0·9) Over time p-value: <0·05	
Mason, 2009 [60]	Craving (ACQ)	Mean score	**Gabapentin** Baseline: not reported Day 7: 40·2 **Placebo** Baseline: not reported Day 7: 33·0 Difference between treatments: 7·2, t=1·08, p>0·05	**Other outcome measures:** Mood (BDI)
	Sleep quality (PSQI)	Mean score	**Gabapentin** Baseline: not reported Day 7: 4·13 **Placebo** Baseline: not reported	

First author, year	Outcome measure	Measure	Estimates	Other comments
			Day 7: 6·50 Difference between treatments: -2·38, t=2·01, p<0·05	
Mason, 2014 [61]	Craving (ACQ)	Baseline: mean score (SD)	**Gabapentin, 900 mg/d** Baseline: 42·5 (12·0) **Gabapentin, 1800 mg/d** Baseline: 42·5 (10·6) **Placebo** Baseline: 42·5 (13·6)	**Comment:** Gabapentin showed significant linear dose effects on craving. Over the course of treatment, significant dose-dependent reductions were obtained on the ACQ (F_2 = 3.56; P = .03; gabapentin, 1800 mg, vs placebo: −6.8 [95% CI, −1.5 to −12.1];t = −2.52;P = .01)
	Sleep quality (PSQI)	Baseline: mean score (SD)	**Gabapentin, 900 mg/d** Baseline: 3·9 (2·3) **Gabapentin, 1800 mg/d** Baseline: 3·5 (2·6) **Placebo** Baseline: 4·9 (2·7)	**Comment:** Gabapentin showed significant linear dose effects on sleep. Over the course of treatment, significant dose-dependent reductions were obtained on the PSQI total score (F_2 = 136; P < .001; gabapentin, 1800 mg, vs placebo: −1.5 [95% CI, −2.1 to −0.8]; t = −4.46; P < .001). **Other outcome measures:** Mood (BDI-II) 4 measures for alcohol abstinence
Melby, 2019 [62]	Sleeping quantity (self-reported hours)	Mean hours (SD)	**Oxytocin** Baseline: not available Day 3: 5·28 (2·95) **Placebo** Baseline: not available Day 3: 5·92 (3·17) p-value: 0·535	**Other outcome measures:** Symptoms (HSCL-10) 1 measure for alcohol abstinence
Miller, 2017 [63]	Sleep quality (daily sleep diary, PSQI or SPQ)	Weighted means (95% CI)	Difference between intervention and control groups: 0·62 (0·28-0·97) Pharmacological effect: 0·43 (0·19-0·67)	**Other outcome measures:** 1 measure for alcohol abstinence 1 measure for depressive symptoms
Minozzi, 2018 [64]	Craving (not specified)	MD baclofen vs. placebo (95% CI)	MD: 1·38 (1·28-4·03)	**Other outcome measures:** Anxiety Depression 4 measures for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
Mishra, 2010 [65]	Craving (ACQ)	Mean score (SD)	**Active rTMS:** Baseline: 245·16 (23·47) Post-treatment: 43·43 (18·05) 1 month post-treatment: 44·25 (73·76) **Sham rTMS:** Baseline: 244·53 (28·72) Post-treatment: 86·73 (11·15) 1 month post-treatment: 89·93 (103·80)	
Mitchell, 2012 [66]	Craving (OCDS)	Mean scores	**Varenicline** Baseline: 12·0 **Placebo** Baseline: 11·0 No exact scores are available at follow-up	**Comment:** Varenicline-treated subjects reported less craving ($\chi 2=13.53$, $p<0.09$, n=35) **Other outcome measures:** 2 measure for smoking dependence 2 measure for alcohol abstinence
Morley, 2006 [67]	Craving (PACS)	Mean score (SD)	**Acamprosate** Baseline: 17·9 (7·3) Week 12: 10·9 (8·5) **Naltrexone** Baseline: 17·8 (7·7) Week 12: 11·1 (8·6) **Placebo** Baseline: 17·6 (6·5) Week 12: 10·9 (8·6)	**Other outcome measures:** 7 measures for alcohol abstinence 2 measures for blood biomarkers
Morley, 2014 [68]	Craving (OCDS-Obsessive)	Mean score (95% CI)	**Baclofen 30 mg** Baseline: 9·13 (6·54-11·71) Week 12: 4·08 (1·63-6·52) **Baclofen 60 mg** Baseline: 5·44 (3·90-6·99) Week 12: 4·47 (2·53-6·42) **Placebo** Baseline: 7·00 (4·10-9·90) Week 12: 4·66 (2·20-7·12) Treatment differences at baseline: p-value=0·05	**Comment:** OCDS total scores are provided only at baseline. Thus, OCDS Obsessive and OCDS Compulsive scores are reported. **Other outcome measures:** 4 measures for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
	Craving (OCDS-Compulsive)	Mean score (95% CI)	**Baclofen 30 mg** Baseline: 16·63 (13·87-19·83) Week 12: 6·93 (2·67-11·19) **Baclofen 60 mg** Baseline: 12·00 (8·90-15·10) Week 12: 8·22 (4·87-11·56) **Placebo** Baseline: 12·50 (7·50-17·50) Week 12: 6·98 (2·70-11·26) Treatment differences at baseline: p-value=0·10	
Muhonen, 2008 [69]	Craving (OCDS)	Mean score (SD)	**Memantine** Baseline: 18·8 (6·9) Week 26: 10·6 (7·2) **Escitalopram** Baseline: 20·4 (4·9) Week 26: 12·8 (8·6) Reduction in both groups combined: F=25·76, p<0·0001 Treatment interaction with time: F=0·69, p=0·56	**Other outcome measures:** 4 measures for alcohol abstinence
Muller, 2015 [70]	Craving (OCDS)	Mean score (SD)	**Baclofen** Baseline: 20·0 (9·0) Week 24: 9·3 (8·8) **Placebo** Baseline: 21·9 (8·0) Week 24: 2·3 (3·2) Treatment difference at baseline: p-value=0·479 Treatment difference at follow-up: p-value=0·294	**Comment:** Information on obsessive and compulsive components of OCDS are also available. **Other outcome measures:** 5 measures for alcohol abstinence Anxiety (HAM-A) Depression (HAM-D)
	Craving (VASC)	Mean score (SD)	**Baclofen** Baseline: 7·9 (12·7) Week 24: 6·7 (9·2) **Placebo** Baseline: 10·3 (16·9) Week 24: 7·0 (6·6)	

First author, year	Outcome measure	Measure	Estimates	Other comments
			Treatment difference at baseline: p-value=0·413 Treatment difference at follow-up: p-value=0·773	
Myrick, 2007 [71]	Sleep quality (ESS)	Mean score (SD)	**Gabapentin** Baseline: 7·5 (4·0) Day 6: 5·9 (3·6) **Placebo** Baseline: 7·3 (2·8) Day 6: 5·9 (2·9) Between group differences p-value: 0·89	
	Craving (AUQ)	Scores	No exact scores are available	**Comment:** There was a non-significant reduction in craving for alcohol on the AUQ scale during the free-choice drinking period in the gabapentin group compared with the placebo group during the first hour - 20.6 versus 26.8 (F=1.946, p=0.178) and during the second hour -16.8 versus 24.4 (F=2.146, p=0.159).
				Other outcome measures: Mood (POMS) 2 measure for alcohol effects
Nava, 2006 [72]	Craving (ACS)	Mean score (SD)	**G-hydroxybutyrate** Baseline: 8·2 (1·3) Month 12: 1·7 (0·7) **Naltrexone** Baseline: 7·3 (1·1) Month 12: 3·6 (1·0) **Disulfiram** Baseline: 7·8 (0·9) Month 12: 3·1 (0·8)	**Other outcome measures:** 1 measure for alcohol abstinence 4 measures for blood biomarkers
Neu, 2018 [73]	Sleep quality (PSQI)	Mean scores (SD)	Admission: 9·1 (4·6) Discharge: 7·1 (3·6) Difference: 2·0 (4·8), p-value=0·0006	
Paparrigopoulos,	Craving (OCDS)	Mean scores (SD)	**Tiagabine**	**Other outcome measures:**

First author, year	Outcome measure	Measure	Estimates	Other comments
2010 [74]			Baseline: 43·6 (3·9) Discharge: 22·1 (3·7) 3rd month: 14·8 (2·9) 6th month: 8·6 (3·1) **Placebo** Baseline: 42·2 (2·8) Discharge: 28·7 (3·2) 3rd month: 20·6 (3·2) 6th month: 15·5 (2·7)	Depression (HDRS) Anxiety (HARS) Overall functioning (GAS) 1 measure for alcohol abstinence 4 measures for blood biomarkers
Paparrigopoulos, 2011 [75]	Craving (OCDS)	Mean scores (SD)	**Topiramate** Baseline: 37·6 (7·8) Day 21: 15·3 (3·9) Day 42: 10·3 (3·1) **Placebo** Baseline: 37·2 (8·3) Day 21: 17·3 (3·9) Day 42: 13·3 (2·8)	**Other outcome measures:** Depression (HDRS) Anxiety (HARS) Overall functioning (GAS) 2 measures of alcohol abstinence
Pedersen, 2013 [76]	Craving (PACS)	Mean scores (SD)	**Oxytocin** Day 1: not reported Day 2:14·6 (5·8) Day 3: 10·1 (7·8) **Placebo** Day 1: not reported Day 2: 21·0 (4·6) Day 3: 12·8 (3·0) Treatment p-value day 2: 0·90 Treatment p-value day 3: 0·544	**Other outcome measures:** 2 measures for alcohol abstinence Mood (POMS)
	Craving (ACVAS)	Mean scores (SD)	**Oxytocin** Day 1: not reported Day 2: 34·9 (20·9) Day 3: 16·7 (18·8) **Placebo** Day 1: not reported Day 2: 86·1 (14·5) Day 3: 38·9 (39·3) Treatment p-value day 2: 0·002 Treatment p-value day 3: 0·2283	
Perney, 2012 [77]	Sleep quantity (SAEI)	Mean score (95% CI)	**Acamprosate**	

First author, year	Outcome measure	Measure	Estimates	Other comments
		at baseline; percentage (%) of mean change over baseline (95% CI)	Baseline: 2·80 (2·52-3·09) Change over time: -52·1% (-63·6 to -38·9) **Placebo** Baseline: 2·83 (2·48-3·19) Change over time: -27·8% (-48·1 to -7·4) Treatment difference at baseline: -0·029 p-value: 0·89 Treatment difference over time: -23·4% (-45·9 to -1·0) p-value: 0·038	
	Sleep quantity (SSI)	Mean score (95% CI)	**Acamprosate** Baseline: 2·40 (2·15-2·65) Change over time: -39·1% (-47·9 to -30·1) **Placebo** Baseline: 2·32 (2·04-2·60) Change over time: -2·6% (-22·6 to 17·4) Treatment difference at baseline: 0·081 (-0·298 to 0·460) p-value: 0·679 Treatment difference over time: -36·4% (-56·5 to -16·3) p-value: 0·001	
Perney, 2018 [78]	Sleep quantity (SSI)	Baseline: not reported Follow-up: mean change over baseline (95% CI)	**Acamprosate** Baseline: not reported Month 4-12: -45·3% **Placebo** Baseline: not reported Month 4-12: -26·4% Treatment difference: -18·98% (-25·46% to -12·49%), p-value<0·001	**Other outcome measures:** 1 measure for alcohol abstinence
Petit, 2017 [79]	Craving (OCDS)	Mean scores (SD)	Day 1: 19·53 (8·34)	**Comment:**

First author, year	Outcome measure	Measure	Estimates	Other comments
			Day 18: 8·47 (5·59)	Obsessive and Compulsive components of OCDS are also provided **Other outcome measures:** Depression (BDI) Mood state (PANAS) 5 measures for cognitive functions
Petit, 2017 [80]	Craving (OCDS)	Mean scores (SD)	Day 1: 19·29 (6·919) Day 18: 8·46 (5·68)	**Comment:** Obsessive and Compulsive components of OCDS are also provided. OCDS score and sub-scores stratified by sex are also provided. **Other outcome measures:** Depression (BDI) Anxiety (STAI) Mood state (PANAS)
Petrakis, 2018 [81]	Craving (OCDS)	Mean score (SD)	**Mecamylamine** Baseline: 18·90 (9·40) Week 12: 8·69 (7·57) **Placebo** Baseline: 17·00 (8·73) Week 12: 8·03 (7·28)	**Other outcome measures:** 3 measures for alcohol abstinence 1 measure for smoking
Ponizovsky, 2015 [82]	Craving (OCDS)	Mean score (SD)	**Baclofen** Baseline: 28·7 (9·8) Week 6: 17·4 (9·2) Week 12: 11·5 (9·3) Week 26: 11·1 (10·1) Week 52: 10·5 (10·1) **Placebo** Baseline: 28·6 (12·3) Week 6: 15·1 (9·4) Week 12: 12·8 (11·2) Week 26: 7·9 (8·6) Week 52: 7·8 (8·7)	**Comment:** Information on obsessive and compulsive components of OCDS are also available. **Other outcome measures:** Depression (BDI) Quality of Life (Q-LES-Q) Emotional distress (GHQ-12) Self-efficacy (GSES) Perceived social support (MSPSS) 2 measures for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
Ray, 2011 [83]	Craving (PACS, OCDS)	Scores	No exact scores are available	**Comment:** Results revealed a significant effect of medication (F=13.29, p<0.01, partial η2=0.51) on craving measured by the PACS, such that quetiapine-treated individuals reported lower alcohol craving than placebo-treated participants. Post hoc comparisons indicated significant group differences during assessment weeks 2, 4, and 5 with t values of 2.68 (p<0.05), 3.32 (p<0.01), and 3.57 (p<0.01), respectively. In addition, there was a significant medication × trial interaction (F=3.11, p<0.05, partial η2=0.19) with regard to OCDS scores, suggesting that quetiapine reduced alcohol craving across trial, as compared with placebo.
	Sleep quality (PSQI)	Scores	No exact scores are available	**Comment:** Results revealed no significant main effect of quetiapine on sleep quality, assessed by the PSQI, Global Scale (F=2.06, p=0.18, partial η2=0.16).
				Other outcome measures: Subjective intoxification (SHAS) Anxiety (BAI) Depression (BDI-II)
Reynaud, 2017 [84]	Craving (OCDS)	Mean score (SD)	**Placebo** Baseline: 17·4 (7·2) Change (6 months): -7·5 (8·4) **Baclofen** Baseline: 19·4 (6·7) Change (6 months): −11·7 (9·6) **Mean difference [95% CI]** −2·86 [−5·22 ; −0·51]; p=0·017	**Other outcome measures**: Anxiety and depression (HAD) Quality of life (AIQoL9) Clinical global impression (CGI) **Comment**: All other scores improved over time in both groups (decreased for HAD and CGI, and increased for AIQoL9). There was a slightly greater improvement for baclofen patients, but without reaching statistical significance.
Roberts, 2017 [85]	Craving (AUQ)	Scores	No exact scores are available	**Comment (results are stratified by**

First author, year	Outcome measure	Measure	Estimates	Other comments
				depression status): Within the depression symptoms group, participants receiving placebo reported greater tonic craving than those receiving 1 mg/day varenicline, t(12) = 2.47, p = 0.029, d = 1.43, and 2 mg/day VAR, t(11) = 3.33, p = 0.007, d = 2.00. There was no significant difference in craving between 1 and 2 mg/day varenicline in this group, t(11) = 1.60, p = 0.138, d = 0.96. Among those in the no depression symptoms group, there was no significant difference in tonic craving among those receiving placebo and 1 mg/day varenicline, t(23) = 0.07, p = 0.944, d = 0.03, or 2 mg/day varenicline, t(23) = 1.35, p = 0.191, d = 0.56. There was no significant difference in tonic craving between participants receiving 1 or 2 mg/day varenicline in this group, t(24) = 1.12, p = 0.273, d = 0.47 **Other outcome measures:** 2 measures for alcohol abstinence
Roberts, 2018 [86]	Craving (AUQ)	Scores	No exact scores are available	**Comment**: No significant main effect of treatment
Rose, 2018 [87]	Craving (not specified)	SMD (95% CI)	Difference between treatments: -0·13 (-0·36, 0·09), p-value: 0·24	**Other outcome measures:** 3 measures for alcohol abstinence Depression Anxiety
Rozatkar, 2016 [88]	Craving (percentage)	Total number (%)	**No craving:** Baseline: 1 (0·8) Week 1: 5 (4·4) Week 2: 21 (22·8) Week 3: 19 (32·7) Week 4: 20 (42·5) **Moderate craving:** Baseline: 32 (28·9) Week 1: 83 (73·4) Week 2: 59 (64·1)	**Other outcome measures:** 1 measure for alcohol use

First author, year	Outcome measure	Measure	Estimates	Other comments
			Week 3: 29 (50·0) Week 4: 20 (42·5) **Severe craving:** Baseline: 80 (70·8) Week 1: 25 (22·1) Week 2: 12 (13·0) Week 3: 10 (17·24) Week 4: 7 (14·9)	
Rubio, 2009 [89]	Craving (analogue scale, not specified)	Mean score (SD)	**Topiramate** Baseline: 7·15 (3·80) Week 12: 3·11 (2·00) **Placebo** Baseline: 7·91 (3·38) Week 12: 5·06 (3·10)	**Other outcome measures:** 3 measures for alcohol abstinence 1 measure for blood biomarker 4 measures for behavioural impulsivity 1 measure for impulsivity
Rubio, 2010 [90]	Craving (analogue scale, not specified)	Mean score (SD)	**Zonisiamide** Baseline: 7·45 (1·43) Week 1: 4·50 (1·01) Week 2: 2·65 (0·94) Week 3: 2·35 (1·10) **Diazepam** Baseline: 7·50 (1·40) Week 1: 4·51 (1·11) Week 2: 3·15 (1·02) Week 3: 3·10 (1·30)	**Other outcome measures:** Anxiety (HAM-A) Depression (HDRS) 1 measure for alcohol abstinence
Schlauch, 2013 [91]	Craving (AUQ)	Mean scores (SD)	Baseline: 19·5 (12·1) Month 2: 17·2 (11·7) Month 4: 17·7 (12·2) Month 6: 17·8 (11·6)	**Other outcome measures:** 4 measures for alcohol abstinence Motivation to change (SOCRATES)
Schmidt, 2011 [92]	Craving (OCDS)	Mean scores	**Abstainers at 24 months after discharge:** 6 months: 1·3 12 months: 1·4 **Non-abstainers at 24 months after discharge:** 6 months: 4·1 12 months: 4·0	**Comment:** Obsessive and Compulsive components of OCDS are also provided
Schneekloth, 2012 [93]	Craving (PACS)	Mean scores (SD)	Admission: 12·2 (8·0) Discharge: 4·0 (4·2)	**Other outcome measures:** Depression (BDI) 1 measure for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
Simioni, 2017 [94]	Craving (OCDS)	Mean score (SD)	**Diazepam 10 days** Baseline: 28·2 (6·9) Month 1: 15·2 (2·6) Month 2: 12·0 (2·7) Month 3: 15·9 (3·0) **Diazepam 30 days** Baseline: 28·2 (7·1) Month 1: 2·8 (3·1) Month 2: 5·5 (3·2) Month 3: 4·3 (3·2) Average treatment effect p-value: 0·007	**Other outcome measures:** Anxiety (HAM-A, STAI-A) 5 measures of alcohol abstinence 1 measure for blood biomarkers
Staner, 2006 [95]	Sleep onset latency (SOL)	Mean minutes (SD)	**Acamprosate** Baseline:36·6 (38·8) Day 14: 31·5 (20·2) **Placebo** Baseline: 25·2 (21·3) Day 14: 22·7 (11·3)	**Comment:** Information on sleep architecture and sleep efficiency is available
	Total sleep time (TST)	Mean minutes (SD)	**Acamprosate** Baseline: 410·7 (35·3) Day 14: 402·0 (62·0) **Placebo** Baseline: 414·9 (41·3) Day 14: 398·6 (59·8)	
	Total time awake (TTA)	Mean minutes (SD)	**Acamprosate** Baseline: 64·3 (32·9) Day 14: 76·9 (59·9) **Placebo** Baseline: 60·0 (45·9) Day 14: 84·3 (65·6)	
Steinig, 2011 [96]	Sleep quantity (ESS)	Mean scores (SD)	During detoxification: 5·8 (4·6) During abstinence: 6·1 (4·8)	**Other outcome measures:** Depression (CES-D10)
	Sleep quality (PSQI)	Mean scores (SD)	During detoxification: 12·8 (5·4) During abstinence: 9·4 (4·6)	
Stock, 2013 [97]	Craving (PACS)	Baseline: mean score (SD) Follow-up: model adjusted mean score (95% CI)	**Gabapentin** Baseline: 30·5 (5·0) Days 1-4: 15·66 (12·74-18·58) Days 5-7: 10·74 (7·09-14·39) **Chlordiazepoxide**	**Other outcome measures:** 1 measure for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
			Baseline: 30·0 (5·8) Days 1-4: 17·05 (12·66-21·45) Days 5-7: 16·78 (10·92-22·64) Treatment difference at baseline: p-value 0·85 Treatment difference at days 1-4: -1·39 (-6·48 to 3·70), p-value 0·59 Treatment difference at days 5-7: -6·05 (-12·82 to 0·72), p-value 0·08	
	Daytime sleepiness (ESS)	Baseline: mean score (SD) Follow-up: model adjusted mean score (95% CI)	**Gabapentin** Baseline: 6·2 (6·2) Days 1-4: 5·97 (4·08-7·86) Days 5-7: 2·65 (0·44-4·85) **Chlordiazepoxide** Baseline: 4·7 (4·2) Days 1-4: 5·31 (2·76-7·86) Days 5-7: 6·35 (3·13-9·57) Treatment difference at baseline: p-value 0·47 Treatment difference at days 1-4: 0·66 (-1·93 to 3·26), p-value 0·61 Treatment difference at days 5-7: -3·70 (-7·21 to -0·19), p-value 0·04	
Subbaraman, 2013 [98]	Craving (OCDS)	Mean score (SD)	**Naltrexone** Baseline: 26·4 (8·3) Weeks 0-4: 11·8 (9·0) Weeks 5-12: 10·3 (8·5) Weeks 13-16: 10·8 (9·6) **CBI** Baseline: 27·1 (7·8) Weeks 0-4: 13·5 (8·5) Weeks 5-12: 11·2 (8·5) Weeks 13-16: 10·9 (8·6) **Naltrexone + CBI** Baseline: 26·8 (8·3) Weeks 0-4: 12·3 (8·6)	**Other outcome measures:** 1 measure for alcohol abstinence

First author, year	Outcome measure	Measure	Estimates	Other comments
			Weeks 5-12: 10·8 (9·2) Weeks 13-16: 10·4 (8·9) **Placebo** Baseline: 26·8 (8·4) Weeks 0-4: 13·5 (9·4) Weeks 5-12: 12·7 (9·3) Weeks 13-16: 11·9 (9·1)	
Umut, 2017 [99]	Craving (PACS)	Mean scores (SD)	Initial phase: 25·72 (4·07) Abstinent phase: 3·32 (2·80) Difference over time: p-value<0·0001	**Other outcome measures:** 1 measure for blood biomarkers
	Craving (OCDS)	Mean scores (SD)	Initial phase: 44·48 (7·98) Abstinent phase: 4·86 (4·14) Difference over time: p-value<0·0001	
Von der Goltz, 2011 [100]	Craving (OCDS)	Mean scores (SD)	Day1: 31·59 (1·55) Day 15: 21·81 (1·79) p-value<0·001	**Other outcome measures:** Depression (BSI) Anxiety (BSI) Psychological symptoms (BSI) 3 measures for blood biomarkers
	Craving (VAS)	Mean scores (SD)	Day1: 52·50 (6·12) Day 15: 15·00 (3·66) p-value<0·001	
Wilhelm, 2011 [101]	Craving (OCDS)	Mean scores (SD)	Day 1:19·76 (6·9) Day 7: 10·49 (7·3) Day 14: 9·72 (6·7)	**Other outcome measures:** 1 measure for alcohol dependence Depression (BDI) Anxiety (STAI)
	Craving (PACS)	Mean scores (SD)	Day 1:16·23 (7·4) Day 7: 5·93 (5·5) Day 14: 4·55 (5·5)	
Zgierska, 2008 [102]	Craving (OCDS)	Mean scores (SD)	Baseline: 10·7 (6·4) Week 4: 7·5 (4·9) Week 8: 7·6 (6·6) Week 12 (4 weeks after stopping treatment): 8·1 (7·3) Week 16 (4 weeks after stopping treatment): 6·7 (6·9)	**Other outcome measures:** Depression (SCL-90R) Anxiety (SCL-90R) 3 measures for blood biomarkers 1 measure for stress symptoms 3 measures for alcohol abstinence

ACQ: Alcohol Craving Questionnaire; ACQ-SF-R, Alcohol Craving Questionnaire Short-Form; ACS: Alcohol Craving Scale; ACVAS: Alcohol Craving Visual Analog Scale; A-OCDS: Adolescent Obsessive Compulsive Drinking Scale; AIQoL9: 9-item alcohol dependence quality of life; AUDIT: Alcohol-Use Disorders Identification Test; AUQ: Alcohol Urge Questionnaire; AW: alcohol withdrawal; BDI: Beck Depression Inventory;

BDI-II: Beck Depression Inventory, 2nd edition; BRMS: Bech-Rafaelsen Melancholia Scale; BSI: Brief Symptoms Inventory; CBI: Combined Behavioral Intervention; CBM: Cognitive Bias Modification; CGI: Clinical Global Impression; CI: confidence interval; CIWA-AR: Clinical Institute Withdrawal Assessment for Alcohol Revised; EQ-5D: European Quality of Life Questionnaire; ESS: Epworth Sleepiness Scale; GAS: Global Assessment Scale; GHQ-12: General Health Questionnaire; GSES: General Self-Efficacy Scale; GSI: General Symptoms Index; HAD: Hospital Anxiety and Depression; HAM-A: Hamilton Anxiety Scale; HAM-D: Hamilton Depression Rating Scale; HARS: Hamilton Anxiety Rating Scale; HDRS: Hamilton Depression Rating Scale; HSCL-10: Hopkin Symptom Check List-10; IIT: intention to treat; IQR: interquartile range; ISI: Insomnia Severity Index; MD: mean difference; MSPSS: Multidimensional Scale of Perceived Social Support; OCDS: Obsessive-Compulsive Drinking Scale; PACS: Penn Alcohol Craving Scale; PCL-C: Post-Traumatic Stress Disorders Checklist-Civilian Version; POMS: Profile of Mood States; PP: per protocol; PSQI: Pittsburgh Sleep Quality Index; QL-index: Quality of life index; Q-LES-Q: Quality of Life Employment and Satisfaction Questionnaire; rTMS: repetitive transcranial magnetic stimulation; SAEI: Sleep sum score index; SCL-90-R: Symptom Check-List 90 Revised; SD: standard deviation; SDS: Zung Self-Rating Depression Scale; SE: standard error; SF-12: 12-item Medical Outcomes Study Short Form; SF-36: 36-item Medical Outcomes Study Short Form; SHAPS: Snaith Hamilton Pleasure Scale; SMD: standardized mean difference; SPQ: Sleep Quality Index; STAI: Spielberger Trait and State Anxiety Inventory; STAI-A: State Anxiety Inventory-A; SSI: Short sleep index; tDCS transcranial direct current stimulation; UF: Urge Form; VAS; Visual Analogue Scale; VASa: Visual Analogue Scale for Anhedonia; VASc: Visual Analogue Scale for Craving

Supplementary Table 4: Description of studies providing at least two estimates of measure of **craving** (at baseline and at follow-up).

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
Baltieri, 2008 [4] Brazil	RCT	Male alcohol dependent outpatients Topiramate, 300 mg/d (n=52) vs.Naltrexone, 50 mg/d (n=49) Placebo (n=54)	Craving (OCDS)	Mean score (SD)	**Topiramate** Baseline: 50·7 (13·5) Week 12: 22·4 (9·4) **Naltrexone** Baseline: 50·9 (13·6) Week 12: 22·4 (10·3) **Placebo** Baseline: 47·9 (12·6) Week 12: 21·9 (8·6)
Beraha, 2016 [5] Netherlands	RCT	Alcohol dependent patients randomized after detoxification Baclofen, max 150 mg/d (n=58) vs. Baclofen, max 30 mg/d (n=31) vs Placebo (n=62)	Craving (OCDS)	Mean score (SD)	**Baclofen, 150 mg/d** Baseline: 28·2 (9·2) Week 4: 13·6 (5·8) Week 16: 12·4 (4·9) **Baclofen, 30 mg/d** Baseline: 29·5 (11·1) Week 4: 13·6 (5·8) Week 16: 12·4 (6·4) **Placebo** Baseline: 30·4 (10·1) Week 4: 13·4 (5·1) Week 16: 12·2 (4·3)
Bogenschutz, 2015 [7] USA	Cohort	Alcohol dependent patients Psilocybin (n=10) administered at week 4 and week 8	Craving (PACS)	Mean scores (SD)	Baseline: 16·00 (5·59) Week 4: 14·10 (7·17) Week 5: 11·89 (8·64) Week 8: 11·56 (5·85) Week 9: 10·00 (6·61) Week 12: 12·11 (8·28) Week 24: 13·00 (9·59) Week 36: 8·11 (9·16)
Bogenschutz, 2016 [8] USA	Cohort	Alcohol dependent patients with anxiety disorder symptoms Lorazepam, max 6 mg/d + Disulfiram, max 500 mg/d (n=41)	Craving (PACS)	Mean scores (SD)	Baseline: 19·56 (8·05) Week 1: 10·26 (7·19) Week 2: 9·64 (7·15) Week 4: 9·53 (7·60) Week 6: 7·61 (7·20) Week 8: 6·52 (6·94)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
					Week 10: 5·25 (5·32) Week 12: 5·17 (7·09) Week 16: 6·59 (8·12) Week 28: 11·10 (10·22)
Caputo, 2011 [10] Italy	Cohort	Alcohol dependent patients Sodium oxybate, 50 mg/kg (n=48)	Craving (ACS)	Mean scores (SD)	**Abstinent patients:** **Patients without psychiatric comorbidities:** Baseline: 13·0 (1·6) Week 12: 0·7 (1·4) **Patients with psychiatric comorbidities:** Baseline: 9·8 (5·3) Week 12: 0·6 (0·9) **Non-abstinent patients:** **Patients without psychiatric comorbidities:** Baseline: 12·7 (1·7) Week 12: 5·0 (2·7) **Patients with psychiatric comorbidities:** Baseline: 13·4 (1·8) Week 12: 5·2 (2·9)
Caputo, 2014 [11] Italy	Cohort	Alcohol dependent consecutive patients Sodium oxybate, 50 mg/kg (n=48)	Craving (ACS)	Mean scores (SD)	**Abstinent patients:** **Lesch I typology of alcohol dependence:** Baseline: 10·7 (5·3) Week 12: 1·2 (1·6) **Lesch II typology of alcohol dependence:** Baseline: 10·5 (3·4) Week 12: 1·0 (1·1) **Lesch III typology of alcohol dependence:** Baseline: 11·0 (5·6) Week 12: 0·25 (0·7) **Lesch IV typology of alcohol dependence:** Baseline: 13·0 (1·7) Week 12: 0·0 (0·0) **Non-abstinent patients:** **Lesch I typology of alcohol dependence:** Baseline: 13·6 (0·9) Week 12: 5·0 (2·4) **Lesch II typology of alcohol dependence:** Baseline: 10·0 (5·2) Week 12: 5·0 (1·7) **Lesch III typology of alcohol dependence:**

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
					Baseline: 13·0 (2·4) Week 12: 5·2 (4·4) **Lesch IV typology of alcohol dependence:** Baseline: 13·5 (1·4) Week 12: 5·1 (2·5)
Chakravorty, 2019 [13] USA	RCT	Veterans alcohol dependent patients CMB-I (n=11) vs. Monitor only (n=11)	Craving (PACS)	Baseline: mean scores (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 9·0 (6·3) Week 8 (end of treatment): -0·5 (-3·6;2·6) Month 3 after treatment: -0·6 (-4·1;2·9) Month 6 after treatment: -0·6 (-4·6;3·2) **CBI-I:** Baseline: 6·1 (7·2) Week 8 (end of treatment): -0·7 (-3·9;-2·3) Month 3 after treatment: -0·8 (-4·4;2·6) Month 6 after treatment: -0·9 (-4·9;2·9)
Collins, 2015 [14] USA	Cohort	Homeless alcohol dependent patients Naltrexone (n=31) Harm reduction counselling and medication management	Craving (PACS)	Median scores (IQR)	Baseline: 21·00 (14·00-28·75) Week 12: 14·00 (5·50-24·50)
de Bejczy, 2015 [16] Sweden	RCT	Alcohol dependent patients Varenicline, max 2mg/d (n=51) vs. Placebo (n=64)	Craving (OCDS)	Baseline: mean score (SD) Follow-up: model-based mean score (CI)	**Varenicline** Baseline: 23·7 (0·77) Week 12 ITT: 4·54 (1·16-7·91) Week 12 PP: 5·64 (1·66-9·62) **Placebo** Baseline: 22·3 (0·66) Week 12 ITT: 6·18 (3·04-9·31) Week 12 PP: 7·62 (3·98-11·26)
De Sousa, 2008 [17] India	RCT	Male alcohol dependent patients Disulfiram, 250 mg/d (n=50) vs. Topiramate, 150 mg/d (n=50)	Craving (craving scale, not specified)	Mean score (SD)	**Disulfiram** Baseline: 56 (16) Month 9: 12·7 (9·3) **Topiramate** Baseline: 53 (17) Month 9: 18·2 (9·9)
De Sousa, 2014 [18] India	RCT	Adolescents with alcohol dependence Disulfiram, max 500 mg/d (n=29) vs.	Craving (A-OCDS)	Mean score (SD)	**Disulfiram** Baseline:11·8 (4·3) Month 6: 9·2 (3·1) **Naltrexone**

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
		Naltrexone, 100 mg/d (n=29)			Baseline:11·4 (4·1) Month 6: 5·3 (1·8)
Den Uyl, 2014 [20] Germany	RCT	Alcohol dependent inpatients Active tDCS + CBM (n=20) vs. Sham tDCS + CBM (n=21) vs. Active tDCS separate from CBM (n=21)	Craving (PACS)	Mean score (SD)	**Active tDCS +CBM** Pre-treatment: 7·1 (1·0) Post-treatment: 5·6 (1·0) **Sham tDCS +CBM** Pre-treatment: 5·9 (1·0) Post-treatment: 4·4 (1·0) **Active tDCS + separate CBM** Pre-treatment: 4·4 (1·0) Post-treatment: 4·1 (1·0)
Di Nicola, 2010 [21] Italy	Cohort	Alcohol dependent patients with mild-to-moderate AWS Pregabalin, max 450 mg/d (n=40) Supportive self-help group run by counsellors or psychologists	Craving (VASc)	Mean scores	**Patients who remained alcohol free** Day 1: 3·5 Day 2: 2·3 Day 3: 1·5 Day 4: 1·0 Day 5: 0·7 Day 6: 0·5 Day 7: 0·3 Day 14: 0·6
Florez, 2008 [22] Spain	RCT	Alcohol dependent patients Topiramate, max 400 mg/d (n=51) vs. Naltrexone, 50 mg/d (n=51)	Craving (OCDS)	Mean score (SD)	**Naltrexone** Baseline: 18·33 (7·064) Month 3: 5·65 (8·492) Month 6: 5·11 (6·846) **Topiramate** Baseline: 16·04 (7·150) Month 3: 2·75 (3·928) Month 6: 2·72 (4·595)
Fucito, 2011 [24] USA	RCT	Heavy drinkers Varenicline (active pretratment+active treatment), 2 mg/d (n=15) vs. Placebo (placebo pretreatment+active treatment) (n=15)	Craving (OCDS)	Mean score (SD)	**Varenicline** Baseline: 7·95 (3·71) Week 8 change: 2·43 (3·16) **Placebo** Baseline: 8·16 (3·43) Week 8 change: 1·57 (3·57)
Garbutt, 2010 [26] USA	RCT	Alcohol dependent subjects Baclofen, 30 mg/d	Craving (PACS)	Mean (SD)	**Baclofen** Baseline: 14·4 (5·8) Week 12: 11·6 (5·9)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
		(n=40) vs. Placebo (n=40)			**Placebo** Baseline: 16·4 (5·6) Week 12: 13·5 (6·3)
Gleissenthall, 2014 [27] Austria	Cohort	Alcohol dependent in-patients Medical management + psychoteraphy + physical exercise (n=54)	Craving (OCDS)	Mean scores (SD)	Week 1: 16·7 (7·7) Week 8: 9·1 (6·4)
Gupta, 2017 [29] India	RCT	Alcohol dependent outpatients Baclofen, 30 mg/d (n=80) vs. Benfothiamine (n=75)	Craving (OCDS)	Mean score (SD)	**Baclofen** Baseline: 27·63 (7·42) Decrease week 12: 16·56 (10·5) **Benfothiamine** Baseline: 26·22 (8·31) Decrease Week 12: 7·48 (9·75)
Hallgren, 2018 [30] USA	Cohort	Alcohol dependent subjects Prazosin,16 mg/d (n=78)	Craving (PACS)	Multivariate mean scores (SE)	**Abstinence initiators:** Average craving: 2·73 (0·45) Post-abstinence immediate change: -0·82 (0·28) Post-abstinence change over time (continuous, per week): -0·21 (0·04) **Already abstainers:** Average craving: 0·98 (0·22) Change over time: -0·04 (0·02) **Continued drinkers:** Average craving: 4·61 (0·23) Change over time: -0·10 (0·05)
Hauser, 2017 [31] USA	RCT	Veterans alcohol dependent patients Baclofen, 30 mg/d (n=88) vs.Placebo (n=92)	Craving (OCDS)	Mean score (95% CI)	**Baclofen** Baseline: 20·55 (18·51-22·58) Week 6: 10·65 (8·58-12·73) Week 12: 8·79 (6·71-10·86) **Placebo** Baseline: 19·50 (17·51-21·49) Week 6: 12·13 (10·11-14·14) Week 12: 9·52 (7·51-11·54)
Heberlein, 2011 [32] Germany	Case-cohort	Male alcohol dependent subjects admitted at the hospital for detoxification treatment Clomethiazole + oxcarbazepine (n=33)	Craving (OCDS)	Mean scores (SD)	Day 1: 19·42 (7·39) Day 7: 10·59 (6·99) Day 14: 9·34 (6·73)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
Heberlein, 2016 [33] Germany	Case-cohort	Male alcohol dependent subjects Clomethiazole + oxcarbazepine (n=99)	Craving (OCDS)	Mean scores (SD)	**Total:** Day 1: 19·80 (6·70) Day 7: 10·92 (7·42) Day 14: 9·88 (6·71) **Early abstinent:** Day 1: 18·42 (6·33) Day 7: 8·58 (7·08) Day 14: 8·90 (5·41) **Positive breath alcohol:** Day 1: 19·81 (6·96) Day 7: 11·43 (7·12) Day 14: 10·54 (7·01)
			Craving (PACS)	Mean scores (SD)	**Total:** Day 1: 16·15 (7·56) Day 7: 6·27 (5·78) Day 14: 5·07 (6·17) **Early abstinent:** Day 1: 16·88 (6·30) Day 7: 4·73 (4·52) Day 14: 3·75 (4·09) **Positive breath alcohol:** Day 1: 15·73 (7·75) Day 7: 6·69 (5·95) Day 14: 5·29 (6·48)
Hillemacher, 2006 [34] Germany	Cohort	Smoking alcohol dependent patients or early abstainers hospitalized for detoxification Clomethiazole, max 2549 mg/d + carbamazepine, max 900 mg/d (n=127)	Craving (OCDS)	Mean scores (SD) and median scores (Range)	**Mean scores:** Day 0: 20·7 (7·73) Day 7: 12·3 (6·31) **Median scores:** Day 0: 20·0 (3-36) Day 7: 11·0 (3-33)
Huang, 2014 [35] Taiwan	RCT	Alcohol dependent subjects Dextromethorphan, 360 mg/d (n=18) vs. Placebo (n=22)	Craving (OCDS)	Mean score (SD)	**Dextromethorphan** Day 1: 37·5 (10·9) Day 2: 28·7 (11·5) Day 3: 23·8 (11·9) **Placebo** Day 1: 35·4 (8·1)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
					Day 2: 25·5 (8·7) Day 3: 20·9 (7·6)
Jose, 2019 [36] India	RCT	Alcohol dependent patients Baclofen, max 20 mg/d (n=49) vs. Topiramate, max 100 mg/d (n=45)	Craving (OCDS)	Mean score (SD)	**Baclofen** Baseline: 26·52 (5·581) Week 1: 16·64 (10·123) Week 2: 9·21 (7·942) Week 3: 5·48 (7·983) Week 4: 2·0 (7·036) **Topiramate** Baseline: 28·36 (6·769) Week 1: 26·08 (7·410) Week 2: 19·44 (9·747) Week 3: 14·50 (11·359) Week 4: 6·83 (9·855)
			Craving (VAS)	Mean score (SD)	**Baclofen** Baseline: 81·20 (8·817) Week 1: 63·20 (17·072) Week 2: 39·56 (14·924) Week 3: 24·78 (23·906) Week 4: 6·19 (18·568) **Topiramate** Baseline: 82·40 (10·116) Week 1: 78·00 (11·547) Week 2: 65·20 (17·823) Week 3: 48·63 (26·603) Week 4: 25·56 (28·122)
Kharb, 2018 [39] India	Cohort	Male alcohol dependent patients Lorazepam (n=34)	Craving (PACS)	Mean scores (SD)	**Non-relapsed patients:** At discharge: 7·78 (1·56) 1 month later: 12·44 (4·33) **Relapsed patients:** At discharge: 15·90 (2·51) 1 month later: 22·29 (3·15)
Koethe, 2007 [41] Germany	RCT	Alcohol dependent inpatients Oxcarbazepine, 900 mg/d (n=24) vs. Placebo (n=26)	Craving (VAS)	Mean scores (SD)	**Oxcarbazepine** Baseline: not reported Day 6: 18·2 (20·27) **Placebo** Baseline: not reported Day 6: 13·6 (15·83)
Leggio, 2011 [45]	Cohort	Alcohol dependent	Craving	Mean scores	Before treatment: 5·67 (3·36)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
Italy		patients Metadoxine, max 2000 mg/d (n=94)	(VAS)	(SD)	After treatment: 2·70 (3·42)
Likhitsathian, 2013 [47] Thailand	RCT	Alcohol dependent patients Topiramate, max 300 mg/d (n=53) vs. Placebo (n=53)	Craving (11 point-Lickert type questionnaire)	Mean score (SD)	**Topiramate** Baseline: 0·6 (1·5) Week 4: 1·8 (2·1) Week 8: 2·0 (2·1) Week 12: 2·4 (2·4) **Placebo** Baseline: 0·9 (1·7) Week 4: 1·7 (2·0) Week 8: 2·0 (2·0) Week 12: 2·0 (2·2)
Litten, 2013 [48] USA	RCT	Alcohol dependent patients Varenicline, max 2 mg/d (n=97) vs. Placebo (n=101)	Craving (PACS)	Baseline: mean score (SD) Follow-up: model-based mean score (95% CI)	**Varenicline** Baseline: 17·7 (6·2) Weeks 6, 10, 14: 9·9 (8·9-10·9) **Placebo** Baseline: 16·7 (6·8) Weeks 6, 10, 14: 11·6 (10·7-12·6)
Martinotti, 2007 [51] Italy	RCT	Alcohol dependent patients Oxcarbazepine, max 900 mg/d (n=28) vs. Oxcarbazepine, max 1800 mg/d (n=29) vs. Naltrexone, 50 mg/d (n=27)	Craving (OCDS)	Mean score (SD)	**Oxcarbazepine, 900 mg** Baseline: 16·1 (10·2) Day 45: 9·5 (7·2) Day 90: 7·2 (7·4) **Oxcarbazepine, 1800 mg** Baseline: 22·9 (13·3) Day 45: 13·0 (10·4) Day 90: 6·2 (8·8) **Naltrexone** Baseline: 26·8 (9·7) Day 45: 7·1 (6·0) Day 90: 6·8 (7·7)
			Craving (VAS)	Mean score (SD)	**Oxcarbazepine, 900 mg** Baseline: 2·8 (3·8) Day 45: 3·2 (3·7) Day 90: 1·3 (1·9) **Oxcarbazepine, 1800 mg** Baseline: 3·9 (3·6) Day 45: 2·9 (3·3)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
					Day 90: 2·0 (2·4) **Naltrexone** Baseline: 3·4 (3·3) Day 45: 1·8 (1·9) Day 90: 1·1 (1·5)
Martinotti, 2008 [53] Italy	Cohort	Alcohol dependent patients who committed the goal of total abstinence Pregabalin, max 450 mg/d (n=20) Self-help group run by counsellors or psychologists	Craving (VAS)	Mean score (SD)	Baseline: 6·0 (2·1) Week 2: 4·1 (2·0) Week 8: 2·5 (2·0) Week 16: 1·2 (1·9)
Martinotti, 2008 [54] Italy	Cohort	Alcohol dependent outpatients with a comorbid diagnosis of bipolar disorder, schizoaffective disorder, and/or borderline personality disorder Quetiapine, max 800 mg/d (n=28)	Craving (OCDS)	Mean scores (SD)	Baseline: 18·8 (11·7) Week 16: 10·2 (11·3)
			Craving (VAS)	Mean scores (SD)	Baseline: 3·9 (3·3) Week 16: 2·5 (2·3)
Martinotti, 2010 [56] Italy	RCT	Alcohol dependent patients Pregabalin, max 450 mg/d (n=37) vs. Tiapride, max 800 mg/d (n=37) vs. Lorazepam, max 10 mg/d (n=37)	Craving (OCDS)	Mean score (SD)	**Lorazepam** Baseline: 20·03 (8·71) Day 14: 10·94 (9·4) **Tiapride** Baseline: 18·1 (7·44) Day 14: 11·16 (9·34) **Pregabalin** Baseline: 19·25 (7·09) Day 14: 8·47 (7·71)
Martinotti, 2010 [57] Italy	RCT	Alcohol dependent patients Pregabalin, max 450 mg /d (n=31) vs. Naltrexone, 50 mg/d (n=28)	Craving (OCDS)	Mean score (SD)	**Pregabalin** Baseline: 22·1 (10·2) Week 16: 5·5 (7·2) **Naltrexone** Baseline: 26·8 (9·7) Week 16: 7·1 (6·0)
			Craving	Mean score	**Pregabalin**

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
			(VAS)	(SD)	Baseline: 6·0 (1·6) Week 16: 1·3 (1·3) **Naltrexone** Baseline: 5·4 (1·7) Week 16: 2·8 (1·9)
Martinotti, 2014 [59] Italy	RCT	Alcohol dependent patients Topiramate, 100 mg/d (n=26) vs. Placebo (n=26)	Craving (OCDS)	Mean score (SD)	**Topiramate** Baseline: 15·5 (9·2) Week 6: 0·5 (0·6) **Placebo** Baseline: 12·7 (6·2) Week 6: 5·0 (4·6)
			Craving (VAS)	Mean score (SD)	**Topiramate** Baseline: 3·4 (1·6) Week 6: 0·3 (0·1) **Placebo** Baseline: 3·9 (1·7) Week 6: 1·0 (0·9)
Mason, 2009 [60] USA	RCT	Alcohol dependent paid volunteers Gabapentin, 1200 mg/d vs. Placebo	Craving (ACQ)	Mean score	**Gabapentin** Baseline: not reported Day 7: 40·2 **Placebo** Baseline: not reported Day 7: 33·0
Mishra, 2010 [65] India	RCT	Male alcohol dependent subjects Active rTMS (n=30) vs. Sham rTMS (n=15)	Craving (ACQ)	Mean score (SD)	**Active rTMS:** Baseline: 245·16 (23·47) Post-treatment: 43·43 (18·05) 1 month post-treatment: 44·25 (73·76) **Sham rTMS:** Baseline: 244·53 (28·72) Post-treatment: 86·73 (11·15) 1 month post-treatment: 89·93 (103·80)
Morley, 2006 [67] Australia	RCT	Alcohol dependent patients Acamprosate, 1998 mg/d (n=55) vs. Naltrexone, 50 mg/d (n=53) vs. Placebo (n=61)	Craving (PACS)	Mean score (SD)	**Acamprosate** Baseline: 17·9 (7·3) Week 12: 10·9 (8·5) **Naltrexone** Baseline: 17·8 (7·7) Week 12: 11·1 (8·6) **Placebo** Baseline: 17·6 (6·5)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
					Week 12: 10·9 (8·6)
Morley, 2014 [68] Australia	RCT	Alcohol dependent patients Baclofen, 30 mg/d (n=14) vs. Baclofen, 60 mg/d (n=14) vs. Placebo (n=14)	Craving (OCDS-Obsessive)	Mean score (95% CI)	**Baclofen 30 mg** Baseline: 9·13 (6·54-11·71) Week 12: 4·08 (1·63-6·52) **Baclofen 60 mg** Baseline: 5·44 (3·90-6·99) Week 12: 4·47 (2·53-6·42) **Placebo** Baseline: 7·00 (4·10-9·90) Week 12: 4·66 (2·20-7·12)
			Craving (OCDS-Compulsive)	Mean score (95% CI)	**Baclofen 30 mg** Baseline: 16·63 (13·87-19·83) Week 12: 6·93 (2·67-11·19) **Baclofen 60 mg** Baseline: 12·00 (8·90-15·10) Week 12: 8·22 (4·87-11·56) **Placebo** Baseline: 12·50 (7·50-17·50) Week 12: 6·98 (2·70-11·26)
Muhonen, 2008 [69] Finland	RCT	Alcohol dependent patients with depression Memantine, 20 mg/d (n=40) vs. Escitalopram, 20 mg/d (n=40)	Craving (OCDS)	Mean score (SD)	**Memantine** Baseline: 18·8 (6·9) Week 26: 10·6 (7·2) **Escitalopram** Baseline: 20·4 (4·9) Week 26: 12·8 (8·6)
Muller, 2015 [70] Germany	RCT	Alcohol dependent patients Baclofen, max 270 mg/d (n=28) vs. Placebo (n=28)	Craving (OCDS)	Mean score (SD)	**Baclofen** Baseline: 20·0 (9·0) Week 24: 9·3 (8·8) **Placebo** Baseline: 21·9 (8·0) Week 24: 2·3 (3·2)
			Craving (VASC)	Mean score (SD)	**Baclofen** Baseline: 7·9 (12·7) Week 24: 6·7 (9·2) **Placebo** Baseline: 10·3 (16·9) Week 24: 7·0 (6·6)
Nava, 2006 [72] Italy	RCT	Alcohol dependent outpatients	Craving (ACS)	Mean score (SD)	**G-hydroxybutyrate** Baseline: 8·2 (1·3)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
		G-hydroxybutyrate, 50 mg/kg (n=28) vs. Naltrexone, 50 mg/d (n=27) vs. Disulfiram, 200 mg/d (n=31)			Month 12: 1·7 (0·7) **Naltrexone** Baseline: 7·3 (1·1) Month 12: 3·6 (1·0) **Disulfiram** Baseline: 7·8 (0·9) Month 12: 3·1 (0·8)
Paparrigopoulos, 2010 [74] Greece	RCT	Alcohol dependent inpatients Tiagabine, max 20 mg/d (n=60) vs. Placebo (n=60)	Craving (OCDS)	Mean scores (SD)	**Tiagabine** Baseline: 43·6 (3·9) Discharge: 22·1 (3·7) 3rd month: 14·8 (2·9) 6th month: 8·6 (3·1) **Placebo** Baseline: 42·2 (2·8) Discharge: 28·7 (3·2) 3rd month: 20·6 (3·2) 6th month: 15·5 (2·7)
Paparrigopoulos, 2011 [75] Greece	RCT	Alcohol dependent patients Topiramate, max 75 mg/d (n=30) vs. Placebo (n=60)	Craving (OCDS)	Mean scores (SD)	**Topiramate** Baseline: 37·6 (7·8) Day 21: 15·3 (3·9) Day 42: 10·3 (3·1) **Placebo** Baseline: 37·2 (8·3) Day 21: 17·3 (3·9) Day 42: 13·3 (2·8)
Pedersen, 2013 [76] USA	RCT	Alcohol dependent inpatients Intranasal oxytocin, 24 IU/dose (n=7) vs. Placebo (n=4)	Craving (PACS)	Mean scores (SD)	**Oxytocin** Day 1: not reported Day 2:14·6 (5·8) Day 3: 10·1 (7·8) **Placebo** Day 1: not reported Day 2: 21·0 (4·6) Day 3: 12·8 (3·0)
			Craving (ACVAS)	Mean scores (SD)	**Oxytocin** Day 1: not reported Day 2: 34·9 (20·9) Day 3: 16·7 (18·8) **Placebo** Day 1: not reported

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
					Day 2: 86·1 (14·5) Day 3: 38·9 (39·3)
Petit, 2017 [80] Belgium	Cohort	Alcohol dependent patients Diazepam, 29 mg/d (n=41)	Craving (OCDS)	Mean scores (SD)	Day 1: 19·53 (8·34) Day 18: 8·47 (5·59)
Petit, 2017 [79] Belgium	Cohort	Alcohol dependent subjects Diazepam, max 60 mg/d (n=256)	Craving (OCDS)	Mean scores (SD)	Day 1: 19·29 (6·919) Day 18: 8·46 (5·68)
Petrakis, 2018 [81] USA	RCT	Alcohol dependent outpatients Mecamylamine, 10 mg/d (n=65) vs. Placebo (n=63)	Craving (OCDS)	Mean score (SD)	**Mecamylamine** Baseline: 18·90 (9·40) Week 12: 8·69 (7·57) **Placebo** Baseline: 17·00 (8·73) Week 12: 8·03 (7·28)
Ponizovsky, 2015 [82] Israel	RCT	Alcohol dependent patients Baclofen, 50 mg/d (n=32) vs. Placebo (n=32)	Craving (OCDS)	Mean score (SD)	**Baclofen** Baseline: 28·7 (9·8) Week 6: 17·4 (9·2) Week 12: 11·5 (9·3) Week 26: 11·1 (10·1) Week 52: 10·5 (10·1) **Placebo** Baseline: 28·6 (12·3) Week 6: 15·1 (9·4) Week 12: 12·8 (11·2) Week 26: 7·9 (8·6) Week 52: 7·8 (8·7)
Reynaud, 2017 [84] France	RCT	Outpatients fully abstinent Baclofen, 180 mg/d (n=99) vs. Placebo (n=91)	Craving (OCDS)	Mean score (SD)	**Placebo** Baseline: 17·4 (7·2) Change (6 months): -7·5 (8·4) **Baclofen** Baseline: 19·4 (6·7) Change (6 months): −11·7 (9·6) **Mean difference [95% CI]** −2·86 [−5·22 ; −0·51]
Rozatkar, 2016 [88] India	Cohort	Alcohol dependent subjects Multiple drugs (n=113)	Craving (percentage)	Total number (%)	**No craving:** Baseline: 1 (0·8) Week 1: 5 (4·4)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
					Week 2: 21 (22·8) Week 3: 19 (32·7) Week 4: 20 (42·5) **Moderate craving:** Baseline: 32 (28·9) Week 1: 83 (73·4) Week 2: 59 (64·1) Week 3: 29 (50·0) Week 4: 20 (42·5) **Severe craving:** Baseline: 80 (70·8) Week 1: 25 (22·1) Week 2: 12 (13·0) Week 3: 10 (17·24) Week 4: 7 (14·9)
Rubio, 2009 [89] Spain	RCT	Male alcohol dependent patients Topiramate (n=31) vs. Placebo (n=32)	Craving (analogue scale, not specified)	Mean score (SD)	**Topiramate** Baseline: 7·15 (3·80) Week 12: 3·11 (2·00) **Placebo** Baseline: 7·91 (3·38) Week 12: 5·06 (3·10)
Rubio, 2010 [90] Spain	RCT	Alcohol dependent patients Zonisamide, max 600 mg/d (n=20) vs. Diazepam, max 130 mg/d (n=20)	Craving (analogue scale, not specified)	Mean score (SD)	**Zonisiamide** Baseline: 7·45 (1·43) Week 1: 4·50 (1·01) Week 2: 2·65 (0·94) Week 3: 2·35 (1·10) **Diazepam** Baseline: 7·50 (1·40) Week 1: 4·51 (1·11) Week 2: 3·15 (1·02) Week 3: 3·10 (1·30)
Schlauch, 2013 [91] USA	Cohort	Alcohol dependent patients with a current schizophrenia-spectrum or bipolar disorder. Not specified (n=278)	Craving (AUQ)	Mean scores (SD)	Baseline: 19·5 (12·1) Month 2: 17·2 (11·7) Month 4: 17·7 (12·2) Month 6: 17·8 (11·6)
Schmidt, 2011 [92] Germany	Cohort	Alcohol dependent inpatients Standard detoxification	Craving (OCDS)	Mean scores	**Abstainers at 24 months after discharge:** 6 months: 1·3 12 months: 1·4

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
		treatment with psychoanalytical and behavioural approaches (n=198)			**Non-abstainers at 24 months after discharge:** 6 months: 4·1 12 months: 4·0
Schneekloth, 2012 [93] USA	Cohort	All alcohol dependent patients Multiple drugs (n=314) Educational and cognitive behavioural groups	Craving (PACS)	Mean scores (SD)	Admission: 12·2 (8·0) Discharge: 4·0 (4·2)
Simioni, 2017 [94] France	RCT	Alcohol dependent outpatients Diazepam 10 days, max 40 mg/d (n=14) vs. Diazepam 30 days, max 40 mg/d (n=12)	Craving (OCDS)	Mean score (SD)	**Diazepam 10 days** Baseline: 28·2 (6·9) Month 1: 15·2 (2·6) Month 2: 12·0 (2·7) Month 3: 15·9 (3·0) **Diazepam 30 days** Baseline: 28·2 (7·1) Month 1: 2·8 (3·1) Month 2: 5·5 (3·2) Month 3: 4·3 (3·2)
Stock, 2013 [97] USA	RCT	Alcohol dependent veteran subjects Gabapentin, max 1200 mg/d (n=17) vs. Chlordiazepoxide, max 100 mg/d (n=9)	Craving (PACS)	Baseline: mean score (SD) Follow-up: model adjusted mean score (95% CI)	**Gabapentin** Baseline: 30·5 (5·0) Days 1-4: 15·66 (12·74-18·58) Days 5-7: 10·74 (7·09-14·39) **Chlordiazepoxide** Baseline: 30·0 (5·8) Days 1-4: 17·05 (12·66-21·45) Days 5-7: 16·78 (10·92-22·64) Treatment difference at baseline: p-value 0·85 Treatment difference at days 1-4: -1·39 (-6·48 to 3·70), p-value 0·59 Treatment difference at days 5-7: -6·05 (-12·82 to 0·72), p-value 0·08
Subbaraman, 2013 [98] USA	RCT	Newly abstinent partecipants Naltrexone, 100 mg/d (n=209) vs. CBI (n=236) Naltrexone, 100 mg/d + CBI (n=213) vs. Placebo	Craving (OCDS)	Mean score (SD)	**Naltrexone** Baseline: 26·4 (8·3) Weeks 0-4: 11·8 (9·0) Weeks 5-12: 10·3 (8·5) Weeks 13-16: 10·8 (9·6) **CBI**

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
		(n=205)			Baseline: 27·1 (7·8) Weeks 0-4: 13·5 (8·5) Weeks 5-12: 11·2 (8·5) Weeks 13-16: 10·9 (8·6) **Naltrexone + CBI** Baseline: 26·8 (8·3) Weeks 0-4: 12·3 (8·6) Weeks 5-12: 10·8 (9·2) Weeks 13-16: 10·4 (8·9) **Placebo** Baseline: 26·8 (8·4) Weeks 0-4: 13·5 (9·4) Weeks 5-12: 12·7 (9·3) Weeks 13-16: 11·9 (9·1)
Umut, 2017 [99] Turkey	Cohort	Alcohol dependent male inpatients admitted at the hospital for detoxification Not specified (n=43)	Craving (PACS)	Mean scores (SD)	Initial phase: 25·72 (4·07) Abstinent phase: 3·32 (2·80)
			Craving (OCDS)	Mean scores (SD)	Initial phase: 44·48 (7·98) Abstinent phase: 4·86 (4·14)
Von der Goltz, 2011 [100] Germany	Cohort	Alcohol dependent inpatients Not specified (n=34)	Craving (OCDS)	Mean scores (SD)	Day 1: 31·59 (1·55) Day 15: 21·81 (1·79)
			Craving (VAS)	Mean scores (SD)	Day 1: 52·50 (6·12) Day 15: 15·00 (3·66)
Wilhelm, 2011 [101] Germany	Case-cohort	Male alcohol dependent patients Not specified (n=99)	Craving (OCDS)	Mean scores (SD)	Day 1:19·76 (6·9) Day 7: 10·49 (7·3) Day 14: 9·72 (6·7)
			Craving (PACS)	Mean scores (SD)	Day 1:16·23 (7·4) Day 7: 5·93 (5·5) Day 14: 4·55 (5·5)
Zgierska, 2008 [102] USA	Cohort	Alcohol dependent patients Course in “Mindfulness Meditation Relapse Prevention” (“meditation”) (n=19)	Craving (OCDS)	Mean scores (SD)	Baseline: 10·7 (6·4) Week 4: 7·5 (4·9) Week 8: 7·6 (6·6) Week 12 (4 weeks after stopping treatment): 8·1 (7·3) Week 16 (4 weeks after stopping treatment): 6·7 (6·9)

ACQ: Alcohol Craving Questionnaire; ACQ-SF-R, Alcohol Craving Questionnaire Short-Form; ACS: Alcohol Craving Scale; ACVAS: Alcohol Craving Visual Analog Scale; A-OCDS: Adolescent Obsessive Compulsive Drinking Scale; AIQoL9: 9-item alcohol dependence quality of life;

AUDIT: Alcohol-Use Disorders Identification Test; AUQ: Alcohol Urge Questionnaire; AW: alcohol withdrawal; BDI: Beck Depression Inventory; BDI-II: Beck Depression Inventory, 2nd edition; BRMS: Bech-Rafaelsen Melancholia Scale; BSI: Brief Symptoms Inventory; CBI: Combined Behavioral Intervention; CBM: Cognitive Bias Modification; CGI: Clinical Global Impression; CI: confidence interval; CIWA-AR: Clinical Institute Withdrawal Assessment for Alcohol Revised; EQ-5D: European Quality of Life Questionnaire; ESS: Epworth Sleepiness Scale; GAS: Global Assessment Scale; GHQ-12: General Health Questionnaire; GSES: General Self-Efficacy Scale; GSI: General Symptoms Index; HAD: Hospital Anxiety and Depression; HAM-A: Hamilton Anxiety Scale; HAM-D: Hamilton Depression Rating Scale; HARS: Hamilton Anxiety Rating Scale; HDRS: Hamilton Depression Rating Scale; HSCL-10: Hopkin Symptom Check List-10; IIT: intention to treat; IQR: interquartile range; ISI: Insomnia Severity Index; MD: mean difference; MSPSS: Multidimensional Scale of Perceived Social Support; OCDS: Obsessive-Compulsive Drinking Scale; PACS: Penn Alcohol Craving Scale; PCL-C: Post-Traumatic Stress Disorders Checklist-Civilian Version; POMS: Profile of Mood States; PP: per protocol; PSQI: Pittsburgh Sleep Quality Index; QL-index: Quality of life index; Q-LES-Q: Quality of Life Employment and Satisfaction Questionnaire; rTMS: repetitive transcranial magnetic stimulation; SAEI: Sleep sum score index; SCL-90-R: Symptom Check-List 90 Revised; SD: standard deviation; SDS: Zung Self-Rating Depression Scale; SE: standard error; SF-12: 12-item Medical Outcomes Study Short Form; SF-36: 36-item Medical Outcomes Study Short Form; SHAPS: Snaith Hamilton Pleasure Scale; SMD: standardized mean difference; SPQ: Sleep Quality Index; STAI: Spielberger Trait and State Anxiety Inventory; STAI-A: State Anxiety Inventory-A; SSI: Short sleep index; tDCS transcranial direct current stimulation; UF: Urge Form; VAS; Visual Analogue Scale; VASa: Visual Analogue Scale for Anhedonia; VASc: Visual Analogue Scale for Craving

Supplementary Table 5: Description of studies providing at least two estimates of measure of **sleep disorder** (at baseline and at follow-up).

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
Brower, 2008 [9] USA	RCT	Alcohol dependent outpatients Gabapentin 1500 mg/d (n=10) vs. Placebo (n=11)	Sleep Problems Questionnaire (SPQ)	Median score (IQR)	**Gabapentin** Baseline: 14·5 (12·0-17·8) 6 weeks: 8·0 (2·5-12·0) 12 weeks: 9·0 (7·0-14·0) **Placebo** Baseline: 18·0 (5·0-19·0) 6 weeks: 8·0 (2·5-11·5) 12 weeks: 2·5 (0·3-7·8)
			Total Sleep Time (TST) measured with polysomnography	Median minutes (IQR)	**Gabapentin** Baseline: 314·5 (294·8-390·8) 3 weeks: 310·9 (281·6-371·0) **Placebo** Baseline: 326·4 (295·8-372·9) 3 weeks: 295·5 (234·4-364·7)
			Sleep onset latency (SOL) measured with polysomnography	Median minutes (IQR)	**Gabapentin** Baseline: 13·8 (6·5-26·1) 3 weeks: 31·0 (20·6-53·6) **Placebo** Baseline: 22·8 (10·9-30·0) 3 weeks: 16·5 (3·9-40·9)
			Wake time after sleep onset (WASO) measured with polysomnography	Median minutes (IQR)	**Gabapentin** Baseline: 70·5 (48·1-84·5) 3 weeks: 76·3 (54·4-94·5) **Placebo** Baseline: 94·5 (65·6-119·3) 3 weeks: 89·8 (51·4-171·8)
Chakravorty, 2014 [12] USA	RCT	Alcohol dependent patients who complained of sleep-related disturbance Quetiapine, max 400 mg/d (n=10) vs. Placebo (n=10)	Wake time after sleep onset (WASO)	Mean minutes (SD)	**Quetiapine** Baseline: 101·07 (81·80) Week 8: 60·71 (71·41) **Placebo** Baseline: 61·17 (31·77) Week 8: 66·99 (60·62)
			Sleep onset latency (SOL)	Mean minutes (SD)	**Quetiapine** Baseline: 25·42 (14·12) Week 8: 37·11 (40·06)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
					Placebo Baseline: 29·04 (29·51) Week 8: 10·81 (9·64)
			Rapid eye movement sleep latency (REM-L)	Mean minutes (SD)	**Quetiapine** Baseline: 108·10 (51·49) Week 8: 115·00 (63·57) **Placebo** Baseline: 92·60 (52·81) Week 8: 88·75 (53·78)
Chakravorty, 2019 [13] USA	RCT	Veterans alcohol dependent subjects with insomnia symptoms CMB-I (n=11) vs. Monitor only (n=11)	Sleep latency (SL)	Baseline: mean minutes (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 33·9 (20·3) Week 8 (end of treatment): 20·6 (-18·2;59·5) Month 3 after treatment: 23·2 (-20·5;67·0) Month 6 after treatment: 25·8 (-22·7;74·4) **CBI-I:** Baseline: 69·0 (65·5) Week 8 (end of treatment): -45·1(-78·0;-12·2) Month 3 after treatment: -36·8 (-75·9;2·3) Month 6 after treatment: -25·4 (-71·9;20·9)
			Insomnia Severity Index (ISI) total score	Baseline: mean scores (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 19·8 (5·6) Week 8 (end of treatment): -8·4 (-16·3;-0·5) Month 3 after treatment: -9·4 (-18·3;-0·5) Month 6 after treatment: -10·5 (-20·4;-0·6) **CBI-I:** Baseline: 18·0 (5·6) Week 8 (end of treatment): -12·7(-15·5;-10·0) Month 3 after treatment: -13·4 (-16·0;-10·9) Month 6 after treatment: -13·9 (-16·3;-11·6)
			Wake time after sleep onset (WASO)	Baseline: mean minutes (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 33·7 (26·4) Week 8 (end of treatment): -1·7 (-17·8;14·3) Month 3 after treatment: -1·9 (-20·1;16·1) Month 6 after treatment: -2·1 (-22·3;17·9) **CBI-I:** Baseline: 43·6 (28·4) Week 8 (end of treatment): -24·5 (-40·8;-8·3) Month 3 after treatment: -27·6 (-45·9;-9·3) Month 6 after treatment: -30·7 (-51·0;-10·4)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
			Total Sleep Time (TST)	Baseline: mean minutes (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 283·1 (115·3) Week 8 (end of treatment): 45·2 (0·8;89·6) Month 3 after treatment: 50·9 (0·9;100·8) Month 6 after treatment: 56·5 (1·0;112·0) **CBI-I:** Baseline: 289·6 (84·7) Week 8 (end of treatment): 102·4 (57·5;147·3) Month 3 after treatment: 115·2 (64·7;165·7) Month 6 after treatment: 128·0 (71·9;184·1)
			Epworth Sleepiness Scale (ESS) total score	Baseline: mean scores (SD) Follow-up: model-estimated change (CI)	**Monitor only:** Baseline: 13·1 (5·6) Week 8 (end of treatment): -0·6 (-2·4;1·0) Month 3 after treatment: -0·7 (-2·7;1·1) Month 6 after treatment: -0·8 (-3·0;1·3) **CBI-I:** Baseline: 9·1 (4·1) Week 8 (end of treatment): -3·1 (-4·9;-1·4) Month 3 after treatment: -3·5 (-5·5;-1·5) Month 6 after treatment: -3·9 (-6·1;-1·7)
Koethe, 2007 [41] Germany	RCT	Alcohol dependent inpatients Oxcarbazepine, 900 mg/d (n=24) vs. Placebo (n=26)	Sleep quality (PSQI)	Mean scores (SD)	**Oxcarbazepine** Baseline: 6·8 (4·17) Day 6: 6·4 (4·81) **Placebo** Baseline: 9·6 (4·02) Day 6: 7·3 (4·45) Treatment difference at the end of the study: -0·9 p-value: 0·54
Lahti, 2011 [43] Finland	RCT	Alcohol dependent patients Nitrous Oxide (n=32) vs. Oxygen (n=30) Air (n=30)	Total sleeping time (TST)	Median hours (SD)	**Nitrous oxide:** Day 1: 6·9 (0·7) Day 2: 6·6 (0·5) Day 3: 6·6 (0·5) **Oxygen:** Day 1: 6·8 (0·3) Day 2: 6·8 (0·5) Day 3: 6·8 (0·5) **Air:** Day 1: 7·1 (0·5) Day 2: 6·6 (0·7)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
					Day 3: 6·8 (0·5)
Mason, 2009 [60] USA	RCT	Alcohol dependent paid volunteers Gabapentin, 1200 mg/d vs. Placebo	Sleep quality (PSQI)	Mean score	**Gabapentin** Baseline: not reported Day 7: 4·13 **Placebo** Baseline: not reported Day 7: 6·50 Difference between treatments: -2·38, t=2·01, p<0·05
Melby, 2019 [62] Norway	RCT	Alcohol dependent patients Oxytocin nasal spray (n=20) vs. Placebo (n=20)	Sleeping quantity (self-reported hours)	Mean hours (SD)	**Oxytocin** Baseline: not available Day 3: 5·28 (2·95) **Placebo** Baseline: not available Day 3: 5·92 (3·17) p-value: 0·535
Myrick, 2007 [71] USA	RCT	Alcohol dependent patients Gabapentin, 1200 mg/d (n=18) vs. Placebo (n=17)	Sleep quality (ESS)	Mean score (SD)	**Gabapentin** Baseline: 7·5 (4·0) Day 6: 5·9 (3·6) **Placebo** Baseline: 7·3 (2·8) Day 6: 5·9 (2·9)
Perney, 2012 [77]	RCT	Alcohol dependent outpatients Acamprosate, 3000 mg/d, (n=335) vs. Placebo (n=257)	Sleep quantity (SAEI)	Mean score (95% CI) at baseline; percentage (%) of mean change over baseline (95% CI)	**Acamprosate** Baseline: 2·80 (2·52-3·09) Change over time: -52·1% (-63·6 to -38·9) **Placebo** Baseline: 2·83 (2·48-3·19) Change over time: -27·8% (-48·1 to -7·4)
			Sleep quantity (SSI)	Mean score (95% CI)	**Acamprosate** Baseline: 2·40 (2·15-2·65) Change over time: -39·1% (-47·9 to -30·1) **Placebo** Baseline: 2·32 (2·04-2·60) Change over time: -2·6% (-22·6 to 17·4)
Staner, 2006 [95] France	RCT	Male alcohol dependent subjects Acamprosate, 666 mg/d (n=12) vs. Placebo (n=12)	Sleep onset latency (SOL)	Mean minutes (SD)	**Acamprosate** Baseline:36·6 (38·8) Day 14: 31·5 (20·2) **Placebo** Baseline: 25·2 (21·3)

First author, year Country	Study design	Population sample size	Outcome (scale)	Measures	Estimate
					Day 14: 22·7 (11·3)
			Total sleep time (TST)	Mean minutes (SD)	**Acamprosate** Baseline: 410·7 (35·3) Day 14: 402·0 (62·0) **Placebo** Baseline: 414·9 (41·3) Day 14: 398·6 (59·8)
			Total time awake (TTA)	Mean minutes (SD)	**Acamprosate** Baseline: 64·3 (32·9) Day 14: 76·9 (59·9) **Placebo** Baseline: 60·0 (45·9) Day 14: 84·3 (65·6)
Steinig, 2011 [96] Germany	Cohort	Alcohol dependent patients Not specified (n=37)	Sleep quantity (ESS)	Mean scores (SD)	During detoxification: 5·8 (4·6) During abstinence: 6·1 (4·8)
			Sleep quality (PSQI)	Mean scores (SD)	During detoxification: 12·8 (5·4) During abstinence: 9·4 (4·6)
Stock, 2013 [97] USA	RCT	Alcohol dependent veteran subjects Gabapentin, max 1200 mg/d (n=17) vs. Chlordiazepoxide, max 100 mg/d (n=9)	Daytime sleepiness (ESS)	Baseline: mean score (SD) Follow-up: model adjusted mean score (95% CI)	**Gabapentin** Baseline: 6·2 (6·2) Days 1-4: 5·97 (4·08-7·86) Days 5-7: 2·65 (0·44-4·85) **Chlordiazepoxide** Baseline: 4·7 (4·2) Days 1-4: 5·31 (2·76-7·86) Days 5-7: 6·35 (3·13-9·57)

ACQ: Alcohol Craving Questionnaire; ACQ-SF-R, Alcohol Craving Questionnaire Short-Form; ACS: Alcohol Craving Scale; ACVAS: Alcohol Craving Visual Analog Scale; A-OCDS: Adolescent Obsessive Compulsive Drinking Scale; AIQoL9: 9-item alcohol dependence quality of life; AUDIT: Alcohol-Use Disorders Identification Test; AUQ: Alcohol Urge Questionnaire; AW: alcohol withdrawal; BDI: Beck Depression Inventory; BDI-II: Beck Depression Inventory, 2nd edition; BRMS: Bech-Rafaelsen Melancholia Scale; BSI: Brief Symptoms Inventory; CBI: Combined Behavioral Intervention; CBM: Cognitive Bias Modification; CGI: Clinical Global Impression; CI: confidence interval; CIWA-AR: Clinical Institute Withdrawal Assessment for Alcohol Revised; EQ-5D: European Quality of Life Questionnaire; ESS: Epworth Sleepiness Scale; GAS: Global Assessment Scale; GHQ-12: General Health Questionnaire; GSES: General Self-Efficacy Scale; GSI: General Symptoms Index; HAD: Hospital Anxiety and Depression; HAM-A: Hamilton Anxiety Scale; HAM-D: Hamilton Depression Rating Scale; HARS: Hamilton Anxiety Rating Scale; HDRS: Hamilton Depression Rating Scale; HSCL-10: Hopkin Symptom Check List-10; IIT: intention to treat; IQR: interquartile range; ISI: Insomnia Severity Index; MD: mean difference; MSPSS: Multidimensional Scale of Perceived Social Support; OCDS: Obsessive-Compulsive Drinking Scale;

PACS: Penn Alcohol Craving Scale; PCL-C: Post-Traumatic Stress Disorders Checklist-Civilian Version; POMS: Profile of Mood States; PP: per protocol; PSQI: Pittsburgh Sleep Quality Index; QL-index: Quality of life index; Q-LES-Q: Quality of Life Employment and Satisfaction Questionnaire; rTMS: repetitive transcranial magnetic stimulation; SAEI: Sleep sum score index; SCL-90-R: Symptom Check-List 90 Revised; SD: standard deviation; SDS: Zung Self-Rating Depression Scale; SE: standard error; SF-12: 12-item Medical Outcomes Study Short Form; SF-36: 36-item Medical Outcomes Study Short Form; SHAPS: Snaith Hamilton Pleasure Scale; SMD: standardized mean difference; SPQ: Sleep Quality Index; STAI: Spielberger Trait and State Anxiety Inventory; STAI-A: State Anxiety Inventory-A; SSI: Short sleep index; tDCS transcranial direct current stimulation; UF: Urge Form; VAS; Visual Analogue Scale; VASa: Visual Analogue Scale for Anhedonia; VASc: Visual Analogue Scale for Craving

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